MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

PROGRAMMABLE TIMER, DIGITAL, KEY CODE HQE

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital programmable timer module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table IV.

| | |
|---|---|
| Name | Programmable timer |
| Family | Digital |
| Key code | HQE |
| Size | 1B |
| Weight | 70 grams maximum |
| Failure rate | 1.659 failures/$10^6$ hours maximum |
| Environment | Class I as specified in MIL-STD-1389 |
| Supply voltage | |
| Pins 1 and 21 | 5.0 Vdc ±10% |
| Maximum power dissipation | |
| 5 volt operation, set | 115 mW (outputs not loaded) |
| Reset | 95 mW (outputs not loaded) |
| 27.5 volt operation, set | 525 mW (outputs not loaded) |
| Reset | 650 mW (outputs not loaded) |
| Regulator voltage | |
| Input (pin 2) | 9.5 Vdc minimum, 27.5 Vdc maximum |
| Output (pin 1 and 21 connected) | 5.0 Vdc ±10 precent |
| Set/Reset input voltage (Note Set/reset input is internally pulled-up to $V_{CC}$ through 4.02 kΩ) | |
| Low level | 0.85 V maximum |
| High level | 3.95 V minimum |
| $C_{IN}$ and P input voltage | |
| Low level | 0.85 V maximum |
| High level | 3.95 V minimum |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Electronic Systems Command, ATTN ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC MISC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| Set/Reset input current | |
| Low level, $V_{IL}$ = 0.0 V, $V_{CC}$ = 5.5 V | -1.25 mA minimum, -1.45 mA maximum |
| High level, $V_{IH}$ = 5.5 V, $V_{CC}$ = 5.5 V | 550 nA maximum |
| $C_{IN}$ and P input current | |
| Low level, $V_{IL}$ = 0.0 V, $V_{CC}$ = 5.5 V | -550 nA maximum |
| High level, $V_{IH}$ = 5.5 V, $V_{CC}$ = 5.5 V | 550 nA maximum |
| Timer 1, D1 output voltage, $V_{CC}$ = 5.0 V | |
| $V_{OL}$ at $I_{OL}$ = 250 µA at 0° and 25°C | 0.5 V maximum |
| $I_{OL}$ = 175 µA at 80°C | |
| $V_{OH}$ at $I_{OH}$ = -150 µA at 0° and 25°C | 4.2 V minimum |
| $I_{OH}$ = -105 µA at 80°C | |
| Timer 2, 3, and 4, D2,D3,D4 and $C_{OUT}$ output voltage, $V_{CC}$ = 5.0 V | |
| $V_{OL}$ at $I_{OL}$ = 100 µA at 0° and 25°C | 0.5 V maximum |
| $I_{OL}$ = 90 µA at 80°C | |
| $V_{OH}$ at $I_{OH}$ = -90 µA at 0° and 25°C | 4.2 V minimum |
| $I_{OH}$ = -65 µA at 80°C | |
| Q and $\overline{Q}$ output voltage, $V_{CC}$ = 5.0 V | |
| $V_{OL}$ at $I_{OL}$ = 3.0 mA at 0° and 25°C | 0.4 V maximum |
| $I_{OL}$ = 2.1 mA at 80°C | |
| $V_{OH}$ at $I_{OH}$ = -450 µA at 0° and 25°C | 4.2 V minimum |
| $I_{OH}$ = -350 µA at 80°C | |
| Open collector transistor condition | |
| Collector - emitter voltage | 30 V maximum |
| Collector leakage current at $V_{CC}$ = 30 V, Vz = 0.4 V | 25 µA maximum |
| Collector - emitter on current | 25 mA maximum |
| Collector - emitter voltage at $I_{CE}$ = 25 mA, Vz = 3.5 V | 0.4 V maximum |
| $C_{IN}$ frequency | |
| at 0° and 25°C | 850 kHz maximum |
| 60°C | 750 kHz maximum |
| $C_{IN}$ pulse width | |
| at 0° and 25°C | 588 ns minimum |
| 60°C | 666 ns minimum |
| $R_1$ pulse width, low level | |
| at 0° and 25°C | 500 ns minimum |
| 60°C | 560 ns minimum |
| $R_2$, $R_3$, and $R_4$ pulse width, low level | |
| at 0° and 25°C | 2.5 µs minimum |
| 60°C | 2.8 µs minimum |
| $C_{IN}$, R, and P input transition time | |
| $t_{THL}$ | 15 µs maximum |
| $t_{TLH}$ | 15 µs maximum |
| Output times | |
| Timer 1· | 125 ±10 ms |
| | 250 ±10 ms |
| | 500 ±10 ms |
| Timer 2· | 500 ±10 ms |
| | 1 sec ±10 ms |
| | 2 sec ±10 ms |
| | 64 sec ±30 ms |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| Timer 3 | 1 sec ±10 ms<br>2 sec ±10 ms<br>16 sec ±20 ms |
| Timer 4 | 4 sec ±10 ms<br>16 sec ±20 ms<br>64 sec ±30 ms |
| Propagation delay time | |
| Timers 1 and 3 | |
| $T_{PLH}$ ($C_L$ = 50 pF) | |
| P to Q | 430 ns |
| P to $\bar{Q}$ | 435 ns |
| Z to T | 1,500 ns |
| $t_{PHL}$ ($C_L$ = 50 pF) | |
| P to Q | 580 ns |
| P to $\bar{Q}$ | 370 ns |
| Z to T | 200 ns |
| Timer 2 | |
| $t_{PLH}$ ($C_L$ = 50 pF) | |
| P to T | 2,100 ns |
| $t_{PHL}$ ($C_L$ = 50 pF) | |
| P to T | 720 ns |
| Timer 4 | |
| $t_{PLH}$ ($C_L$ = 50 pF) | |
| P to T | 1,900 ns |
| $t_{PHL}$ ($C_L$ = 50 pF) | |
| P to T | 610 ns |

1.3 Module function. The programmable timer module designated as key code HQE, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains four independent programmable timers with open collector outputs.

1.3.1 Input to output relationship. Each timer has the input to output relationship as shown in table I with output timer as shown in table II.

1.3 1.1 Functional description. The HQE module contains a power supply voltage regulator, an internal oscillator, and four programmable timer circuits.

1.3.1.1.1 Power supply voltage regulator. The HQE module, using pins 1 and 21, is a 5 volt logic module. However, when a voltage from 9.5 to 27.5 volts is used, the module provides internal voltage regulations. When this voltage is placed on pin 2, pin 1 becomes the output of the power supply voltage regulator. This must be connected to board two, via a pin 1 to pin 21 jumper wire.

1.3.1.1.2 Internal oscillator. The HQE module has an internal 128 Hz (CMOS output) oscillator. The output is internally connected to timer number 1. Timers 2 to 4 must be externally coupled to pin 36 (oscillator output).

1.3.1.1.3 Programmable timers. The HQE module has four independent timers. A timing sequence is started by applying a low-level voltage to the S/R input. The output time is a function of the selection of module program pins as specified in table II. Reset to quiescent state is accomplished by the removal of the low-level voltage (open circuit S/R input) or an application of a high-level voltage to the S/R input. Timers 1 and 3 have output level selection. When pin Q is connected to pin Z, the quiescent state of the output is low. When pin $\bar{Q}$ is connected to pin Z, the quiesent state of the output is high. Timers 2 and 4 are internally connected with $T_2$ in the low-quiescent state and $T_4$ in the high-quiescent state.

1.3.1.2 Modes of operation. The HQE module has three modes of operation, standard programming, addition of standard times and cascading of timers.

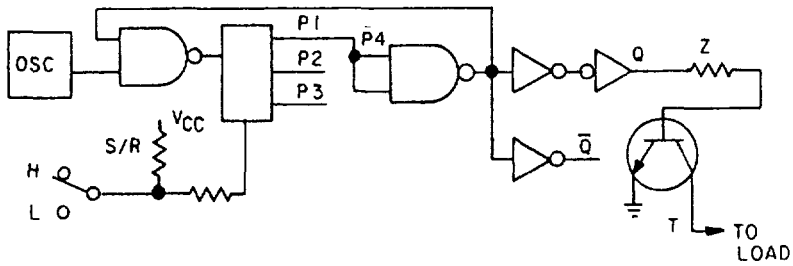


Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE I Input to output relationship.

| Timer 1 and 3 | | | |
|---|---|---|---|
| | Output | | |
| Input S/R | Pin Q to pin Z | Pin $\overline{Q}$ to pin Z | Function |
| H | L | H | Reset |
| ↓ | L | H | Timing initialized |
| L | H | L | Timed out |

| Timer 2 and 4 | | | |
|---|---|---|---|
| | Output | | |
| Input S/R | Timer 2 | Timer 4 | Function |
| H | L | H | Reset |
| ↓ | L | H | Timing initialized |
| L | H | L | Timed out |




NOTES:
1. H = High-level voltage.
2. L = Low-level voltage
3. ↓ = Transition high to low
4. T = Output is an open collector transistor



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE II Timer programming

| Timer No | Time | Binary stage 2n | Connect Pin to pin | | Connect Pin 36 to |
|---|---|---|---|---|---|
| 1 | 125 ms | 5 | 26 | 32 | - |
| 1 | 250 ms | 6 | 27 | 32 | - |
| 1 | 500 ms | 7 | 29 | 32 | - |
| 2 | 500 ms | 7 | 22 | 34 | 33 |
| 2 | 1 sec | 8 | 24 | 34 | 33 |
| 2 | 2 sec | 9 | 25 | 34 | 33 |
| 2 | 64 sec | 14 | 23 | 34 | 33 |
| 3 | 1 sec | 8 | 9 | 20 | 17 |
| 3 | 2 sec | 9 | 8 | 20 | 17 |
| 3 | 16 sec | 12 | 6 | 20 | 17 |
| 4 | 4 sec | 10 | 5 | 19 | 16 |
| 4 | 16 sec | 12 | 4 | 19 | 16 |
| 4 | 64 sec | 14 | 3 | 19 | 16 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

1.3.1.2.1 Standard programming. This mode of operation is shown in table I and table II. With the S/R input in the high state, the timer is disabled and is in a quiescent state. With the application of a low on the S/R input, the timer starts and the output will change state when the programmed time has expired. The state of the output remains constant until the timer is reset.

1.3.1.2.2 Addition of standard time. Standard times, as designated in table II, are added to produce longer times. This is accomplished by connecting two timers in series as shown on figure 1. The output delay time is the sum of timer A plus timer B. Timer A is normally a high-output state so that when timer A times out (output goes low) it sets timer B. The output of timer A can also be used as an output so that sequential timing may be accomplished.

1.3.1.2.3 Cascading of times. Another method of increasing the output time is by cascading two or more timers as shown on figure 2. Timer A becomes an extention of timer B's counter. The output time can be computed by using table III and the following equation:

$$\text{Time out} = \frac{2^{n-1}}{128} \text{ sec}$$

Where n = number of stages used in timers A and B

TABLE III.

| Timer | Output time | Module pin | Equivalent binary counter stage |
|---|---|---|---|
| 1 | 0.125 sec | 26 | 5 |
| 1 | 0.25 sec | 27 | 6 |
| 1 | 0.5 sec | 29 | 7 |
| 2 | 0.5 sec | 22 | 7 |
| 2 | 1 sec | 24 | 8 |
| 2 | 2 sec | 25 | 9 |
| 2 | 64 sec | 23 | 14 |
| 3 | 1 sec | 9 | 8 |
| 3 | 2 sec | 8 | 9 |
| 3 | 16 sec | 6 | 12 |
| 4 | 4 sec | 5 | 10 |
| 4 | 16 sec | 4 | 12 |
| 4 | 64 sec | 3 | 14 |

1.3.2 Program wiring. Module program wiring is restricted to within the module connector only.

1.3.3 Module handling procedures. The HQE module contains COS/MOS devices and are susceptible to damage when exposed to static electrical charge. When handling, shipping, or storing the module, pins should be shorted together with a conductive material or the module should be placed into an electrically conductive bag. The HQE module cannot be plugged into or removed from a hot socket. Failure to comply with the handling and hot socket guidelines could cause irriversible module damage.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

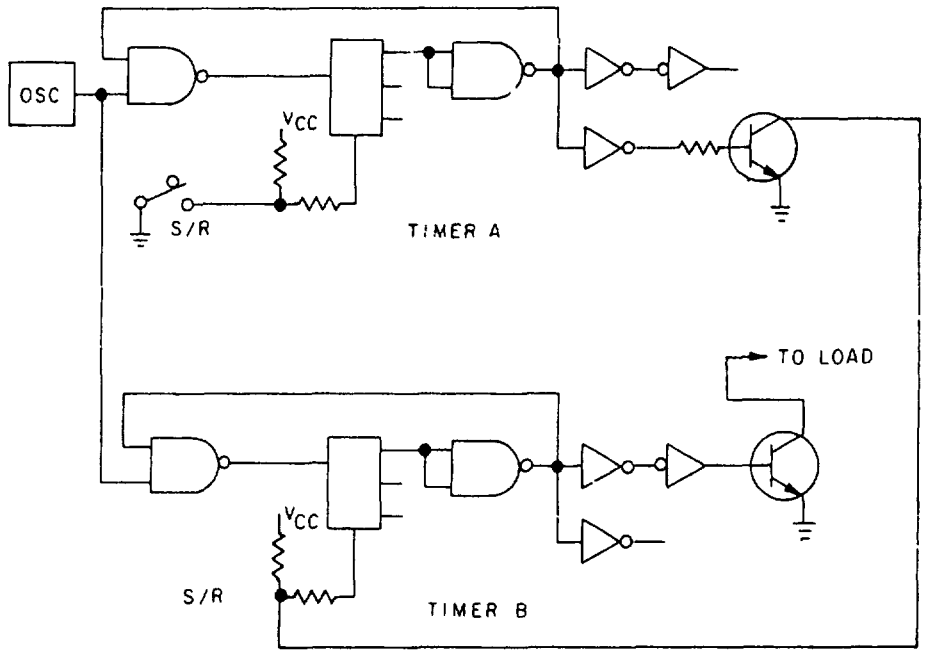



FIGURE 1 Addition of standard time



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

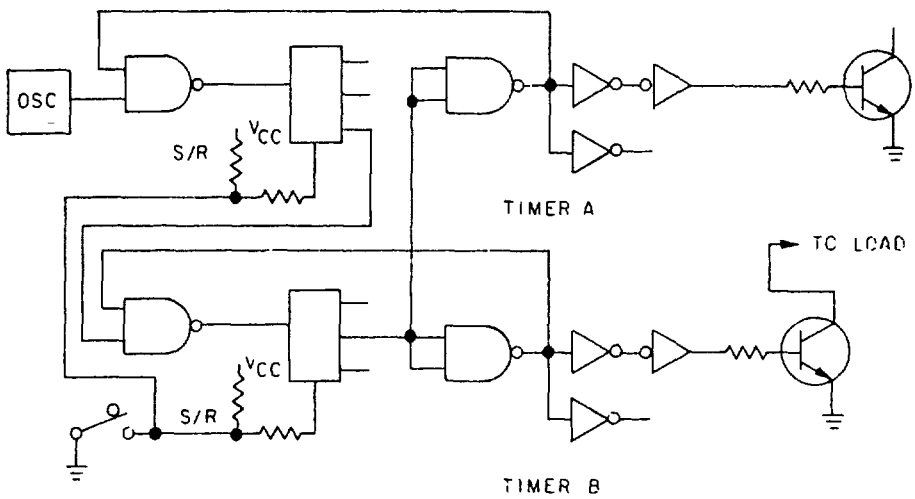



FIGURE 2 Cascading of times



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1 659 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements The module shall conform to the electrical requirements specified in table IV and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be HQE.

3 5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned 1A1 through 1A20 and 1B21 through 1B40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/208, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier)

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 3.

3 7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table IV.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table IV.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table IV for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 Vdc or a power supply voltage from 9.5 Vdc minimum to 27.5 Vdc maximum when the internal voltage regulator is used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below.

| Equipment | MIL-STD-1665 Designator | No. required (maximum per test figure) |
|---|---|---|
| DC power supply | Item P01 | 4 |
| Megohm bridge | Item R06 | 1 |
| Counter | HP5302A, or equivalent | 1 |
| DC voltmeter | Item V03 | 1 |
| Pulse generator | Item J01 | 2 |
| Oscilloscope | Item D05 | 1 |
| Oscilloscope probe | Item Y01 | 2 |
| DC ammeter | Item X01 | 1 |
| Oscilloscope | Item D01 | 1 |
| DC ammeter | Item X03 | 1 |
| Oscilloscope probe | Item Y02 | 1 |
| Oscilloscope | Item D06 | 1 |
| Oscilloscope probe | Item Y04 | 1 |
| Capacitance meter | Item C05 | 1 |
| Capacitance bridge | Item C01 | 1 |
| Resistors | | |
| 1.1 kΩ ±1%, 1 W | | 4 |
| 187 kΩ ±1%, 1/4 W | | 4 |
| 18.2 kΩ ±1%, 1/8 W | | 1 |
| 26.1 kΩ ±1%, 1/8 W | | 1 |
| 45.3 kΩ ±1%, 1/8 W | | 1 |
| 49.9 kΩ ±1%, 1/8 W | | 1 |
| 1.54 kΩ ±1%, 1/8 W | | 1 |
| 2.15 kΩ ±1%, 1/8 W | | 1 |
| 9.31 kΩ ±1%, 1/8 W | | 1 |
| 12.1 kΩ ±1%, 1/8 W | | 1 |
| 28 kΩ ±1%, 1/8 W | | 1 |
| 40.2 kΩ ±1%, 1/8 W | | 1 |
| 51Ω ±5%, 1/4 W | | 2 |
| 46.4 kΩ ±1%, 1/8 W | | 1 |
| 64.9 kΩ ±1%, 1/8 W | | 1 |
| 10 kΩ ±1%, 1/4 W | | 1 |
| 200 kΩ ±1%, 1/8 W | | 1 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

| Capacitors | No. required (maximum per test figure) |
|---|---|
| 50 ±5 pF | 4 |
| Switch | |
| SPST | 1 |

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table IV.

4.4.1 Alternate test methods - Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through CC may be performed in any order with the following exceptions If test conditions A and B are included in the test sequence, the order of testing shall be as follows A, B where applicable followed by the remaining test conditions

4.5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table IV shall be performed in the order specified in 4.4.2 using the initial limits of 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 4

4.5.1.1 Component temperature test The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table IV which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ±5°C ambient

4.5.3 Life tests. The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 4 Following the life test and while still at the maximum class temperature, electrical tests C through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table IV. Following the return of the module to 25°C, electrical tests C through CC shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table IV

5. PACKAGING

5.1 Packaging requirements The packaging requirements shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions Definitions are as specified in MIL-STD-1331

6.2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table IV, the module is considered to have completed its useful life.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE IV Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type (see notes) | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A | Isolation | V | S | 10.0 | ∞ | 1.0 | ∞ | 10.0 | ∞ | 1.0 | ∞ | MΩ |
| B | 7 V $V_{CC}$ stress | VI | S | 7 0 | 9.0 | - | - | - | - | - | - | ms |
| C1-C6 | T output voltage, low level, $V_{CC}$ = 4 5 V | VII | X | 0 0 | 0 4 | 0.0 | 0.4 | 0.0 | 0 4 | 0.0 | 0 4 | V |
| D1-D4 | T output voltage, low level | VIII | | 0 0 | 0 4 | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | |
| D5-D17 | D output voltage, low level | VIII | | 0 0 | 0.5 | 0.0 | 0.5 | 0 0 | 0.5 | 0 0 | 0.5 | |
| D18-D21 | Q and $\bar{Q}$ output voltage, low level | VIII | | 0 0 | 0 4 | 0 0 | 0 4 | 0 0 | 0.4 | 0 0 | 0.4 | |
| D22-D25 | Q and $\bar{Q}$ output voltage, low level | VIII | | 0 0 | 0.2 | 0.0 | 0.2 | 0.0 | 0 2 | 0.0 | 0.2 | |
| E1-E7 | Q, $\bar{Q}$, and D output voltage, high level | IX | | 4 2 | 5 0 | 4.2 | 5 0 | 4 2 | 5 0 | 4.2 | 5 0 | |
| F1-F10 | D output voltage, high level | X | | 4 2 | 5.0 | 4 2 | 5 0 | 4.2 | 5.0 | 4.2 | 5.0 | V |
| G1, G2 | T collector leakage current | XI | | 0 | 25 | 0 | 25 | 0 | 25 | 0 | 25 | µA |
| H1, H2 | T collector leakage current | XII | | 0 | 25 | 0 | 25 | 0 | 25 | 0 | 25 | µA |
| J1 | $C_{OUT}$ voltage,low level | XIII | | 0 0 | 0 4 | 0.0 | 0 4 | 0 0 | 0 4 | 0.0 | 0.4 | V |
| J2 | $C_{OUT}$ voltage, high level | XIII | | 4 2 | 5 0 | 4 2 | 5 0 | 4 2 | 5.0 | 4 2 | 5.0 | V |
| K1-K7 | P, $C_{IN}$, and S/R input current, low level | XIV | | 0 | 55 | 0 | 55 | 0 | 55 | 0 | 55 | nA |
| K8-K11 | P, $C_{IN}$, and S/R input current, low level | XIV | X | 1 25 | 1.40 | 1 25 | 1 40 | 1 25 | 1 40 | 1 25 | 1.40 | mA |

See notes at end of table



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE IV Module electrical requirements - Continued

| Test condition | Parameter | Ref test table | Test type (see notes) | Acceptance requirements 25° +5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| L1, L2 | Z input current, low level | XV | X | 0 | 10 | 0 | 10 | 0 | 10 | 0 | 10 | µA |
| M1-M7 | P, $C_{IN}$ input current, high level | XVI | | 0 | 55 | 0 | 55 | 0 | 55 | 0 | 55 | nA |
| M8-M11 | S/R input current, high level | XVI | | 0 | 200 | 0 | 200 | 0 | 200 | 0 | 200 | nA |
| N1, N2 | Z input current, high level | XVII | | 1 0 | 1 4 | 1 0 | 1 4 | 1 0 | 1 4 | 1 0 | 1 4 | mA |
| P1 | Power supply current | XVIII | | 5 | 20 | 5 | 20 | 5 | 20 | 5 | 20 | mA |
| P2 | ↓ | ↓ | | 7 | 25 | 7 | 25 | 7 | 25 | 7 | 25 | mA |
| P3 | ↓ | ↓ | | 5 | 19 | 5 | 19 | 5 | 19 | 5 | 19 | mA |
| P4 | Power supply current | XVIII | | 7 | 26 | 7 | 26 | 7 | 26 | 7 | 26 | mA |
| Q1, Q2 | Power supply regulation | XIX | | 4 5 | 5 5 | 4 5 | 5 5 | 4 5 | 5 5 | 4 5 | 5 5 | V |
| R | Oscillator repetition period | XX | | 7 81 | 7 82 | 7 81 | 7 82 | 7 81 | 7 82 | 7 81 | 7 82 | ms |
| S1 | D output repetition period | XXI | | 162 | 179 | 162 | 179 | 162 | 179 | 162 | 179 | µs |
| S2 | ↓ | ↓ | | 324 | 359 | 324 | 359 | 324 | 359 | 324 | 359 | µs |
| S3 | ↓ | ↓ | | 649 | 718 | 649 | 718 | 649 | 718 | 649 | 718 | µs |
| S4 | ↓ | ↓ | | 20 79 | 22 98 | 20 79 | 22 98 | 20 79 | 22 98 | 20 79 | 22 98 | ms |
| S5 | ↓ | ↓ | | 649 | 718 | 649 | 718 | 649 | 718 | 649 | 718 | µs |
| S6 | ↓ | ↓ | | 5 19 | 5 79 | 5 19 | 5 79 | 5 19 | 5 79 | 5.19 | 5.79 | ms |
| S7 | D output repetition period | XXI | X | 324 | 359 | 324 | 359 | 324 | 359 | 324 | 359 | µs |

See notes at end of table





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

| Test condition | Parameter | Ref test table | Test type (see notes) | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| S8 | D output repetition period | XXI | X | 1 29 | 1 44 | 1 29 | 1.44 | 1 29 | 1 44 | 1 29 | 1.44 | ms |
| S9 | D output repetition period | XXI | | 5 19 | 5 79 | 5 19 | 5 79 | 5 19 | 5.79 | 5.19 | 5.79 | ms |
| S10 | D output repetition period | XXI | | 20.79 | 22.98 | 20 79 | 22 98 | 20 79 | 22.98 | 20.79 | 22.98 | ms |
| T1 | D output time interval | XXII | | 115 | 135 | 115 | 135 | 115 | 135 | 115 | 135 | ms |
| T2 | D output time interval | XXII | | 240 | 260 | 240 | 260 | 240 | 260 | 240 | 260 | ms |
| T3 | D output time interval | XXII | | 490 | 510 | 490 | 510 | 490 | 510 | 490 | 510 | ms |
| U1-U4 | Timer disable | XXIII | X | 4 5 | 5 0 | 4 5 | 5 0 | 4 5 | 5 0 | 4 5 | 5 0 | V |
| V1-V10 | Timer 2-4 output transition time ($t_{THL}$) | XXIV | S | 2 | 1,500 | 2 | 1,500 | 2 | 1,500 | 2 | 1,500 | ns |
| V11-V20 | Timer 2-4 output transition time ($t_{TLH}$) | XXIV | | 2 | 1,500 | 2 | 1,500 | 2 | 1,500 | 2 | 1,500 | ns |
| W1-W3 | Timer 2-4 reset propagation delay time ($t_{PHL}$) | XXV | | 0 0 | 4 3 | 0.0 | 4 3 | 0 0 | 4 3 | 0.0 | 4.3 | µs |
| X1, X2 | Q output propagation delay time ($t_{PHL}$) | XXVI | | 2 | 580 | 2 | 580 | 2 | 580 | 2 | 580 | ns |
| X3, X4 | $\bar{Q}$ output propagation delay time ($t_{PLH}$) | | | | 435 | | 435 | | 435 | | 435 | |
| X5, X6 | Q output propagation delay time ($t_{PLH}$) | | | | 430 | | 430 | | 430 | | 430 | |
| X7, X8 | $\bar{Q}$ output propagation delay time ($t_{PHL}$) | XXVI | S | 2 | 370 | 2 | 370 | 2 | 370 | 2 | 370 | ns |

See notes at end of table



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

Table IV Module electrical requirements - Continued

| Test condition | Parameter | Ref test table | Test type (see notes) | Acceptance requirements 25° +5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| X9-X12 | Q, $\bar{Q}$ output transition time ($t_{THL}$) | XXVI | S | 2 | 85 | 2 | 85 | 2 | 85 | 2 | 85 | ns |
| X13-X16 | Q, $\bar{Q}$ output transition time ($t_{TLH}$) | XXVI | S | | 390 | | 390 | | 390 | | 390 | |
| Y1, Y2 | T1 and T3 propagation delay time ($t_{PHL}$) | XXVII | | | 200 | | 200 | | 200 | | 200 | |
| Y3, Y4 | T1 and T3 propagation delay time ($t_{PLH}$) | | | | 1,500 | | 1,500 | | 1,500 | | 1,500 | |
| Y5 | T2 output propagation delay time ($t_{PLH}$) | | | | 2,100 | | 2,100 | | 2,100 | | 2,100 | |
| Y6 | T4 output propagation delay time ($t_{PLH}$) | | | | 1,900 | | 1,900 | | 1,900 | | 1,900 | |
| Y7 | T2 output propagation delay time ($t_{PHL}$) | | | | 720 | | 720 | | 720 | | 720 | |
| Y8 | T4 output propagation delay time ($t_{PHL}$) | | | | 610 | | 610 | | 610 | | 610 | |
| Y9, Y10 | T output transition time ($t_{THL}$) | | | | 200 | | 200 | | 200 | | 200 | |
| Y11, Y12 | T output transition time ($t_{TLH}$) | | | | 400 | | 400 | | 400 | | 400 | |
| Y13, Y14 | T output transition time ($t_{THL}$) | | | | 200 | | 200 | | 200 | | 200 | |
| Y15, Y16 | T output transition time ($t_{TLH}$) | XXVII | S | 2 | 400 | 2 | 400 | 2 | 400 | 2 | 400 | ns |

See notes at end of table





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE IV Module electrical requirements - Continued

| Test condition | Parameter | Ref test table | Test type (see notes) | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Z1-Z12 | Output crosstalk, low level | XXVIII | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| Z13-Z24 | Output crosstalk, high level | XXVIII | | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| AA | Power supply transient amplitude | XXIX | | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | mV |
| BB1, BB2 | Power supply filter capacitance | XXX | | 0 01 | 0.04 | 0.01 | 0 04 | 0 01 | 0 04 | 0 01 | 0 04 | µF |
| CC1-CC3 | Input capacitance | XXXI | | 2 | 12 | 2 | 12 | 2 | 12 | 2 | 12 | pF |
| CC4-CC11 | Input capacitance | XXXI | | 2 | 20 | 2 | 20 | 2 | 20 | 2 | 20 | pF |
| CC12, CC13 | Input capacitance | XXXI | Q | 2 | 40 | 2 | 40 | 2 | 40 | 2 | 40 | pF |

NOTES

1 (X) parameters shall be 100 percent tested during module acceptance tests.
2 (S) parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C
3 (Q) parameters shall be tested during qualification testing only



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

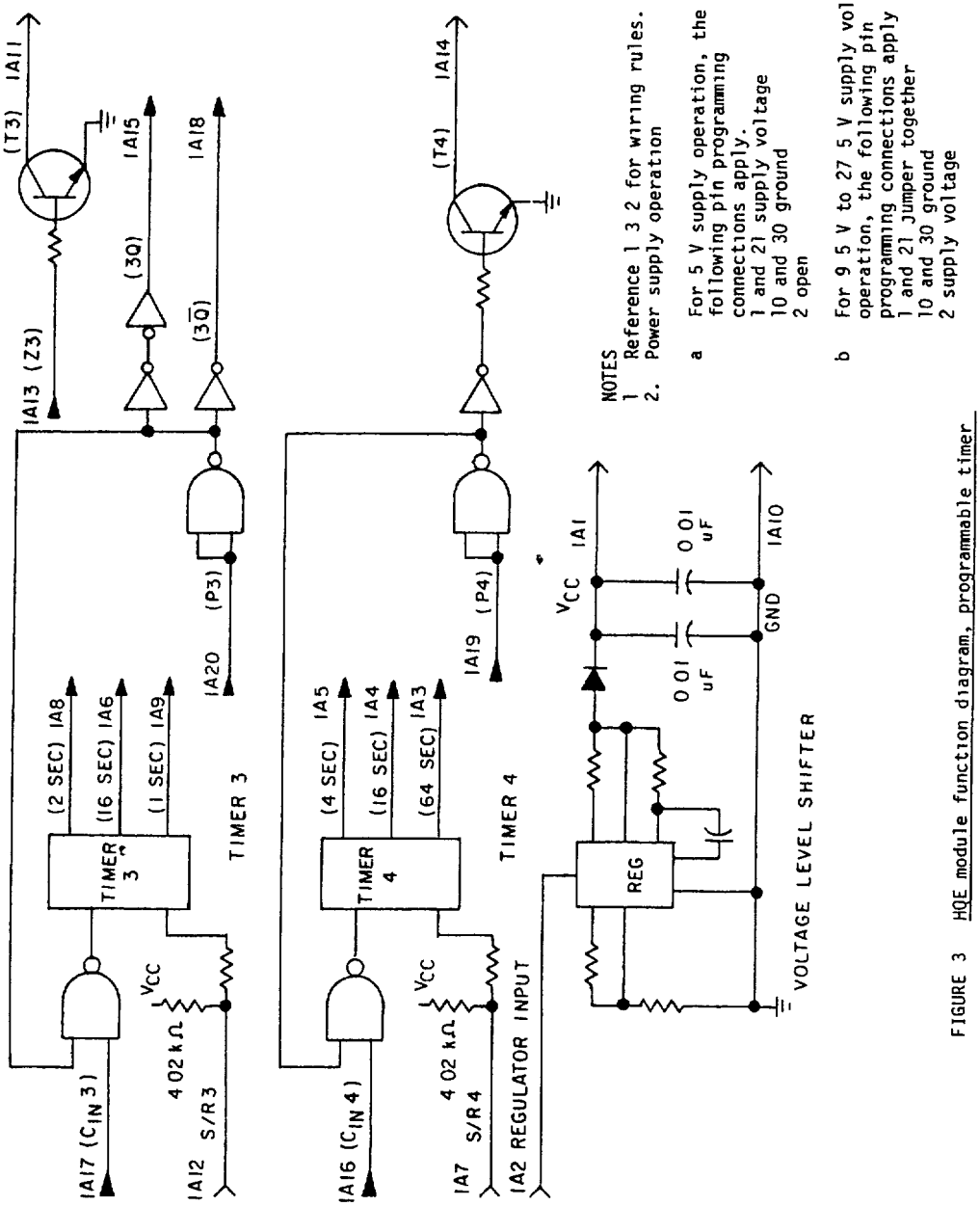
Preparing activity
Navy - EC

Agent
DLA - ES

(Project MISC-N890-82)



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1. Reference 1 3 2 for wiring rules.
2. Power supply operation

a For 5 V supply operation, the following pin programming connections apply.
1 and 21 supply voltage
10 and 30 ground
2 open

b For 9 5 V to 27 5 V supply voltage operation, the following pin programming connections apply
1 and 21 jumper together
10 and 30 ground
2 supply voltage

FIGURE 3 HQE module function diagram, programmable timer



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

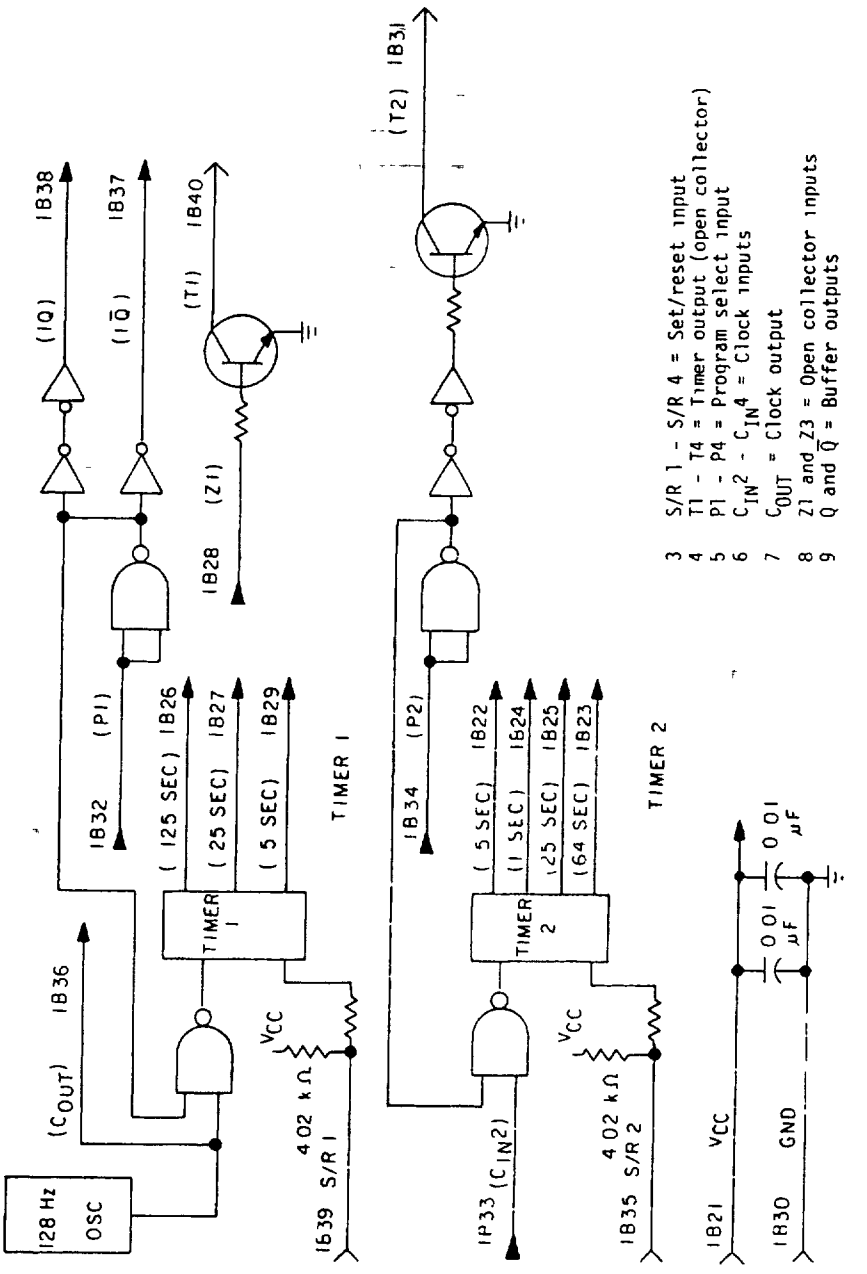
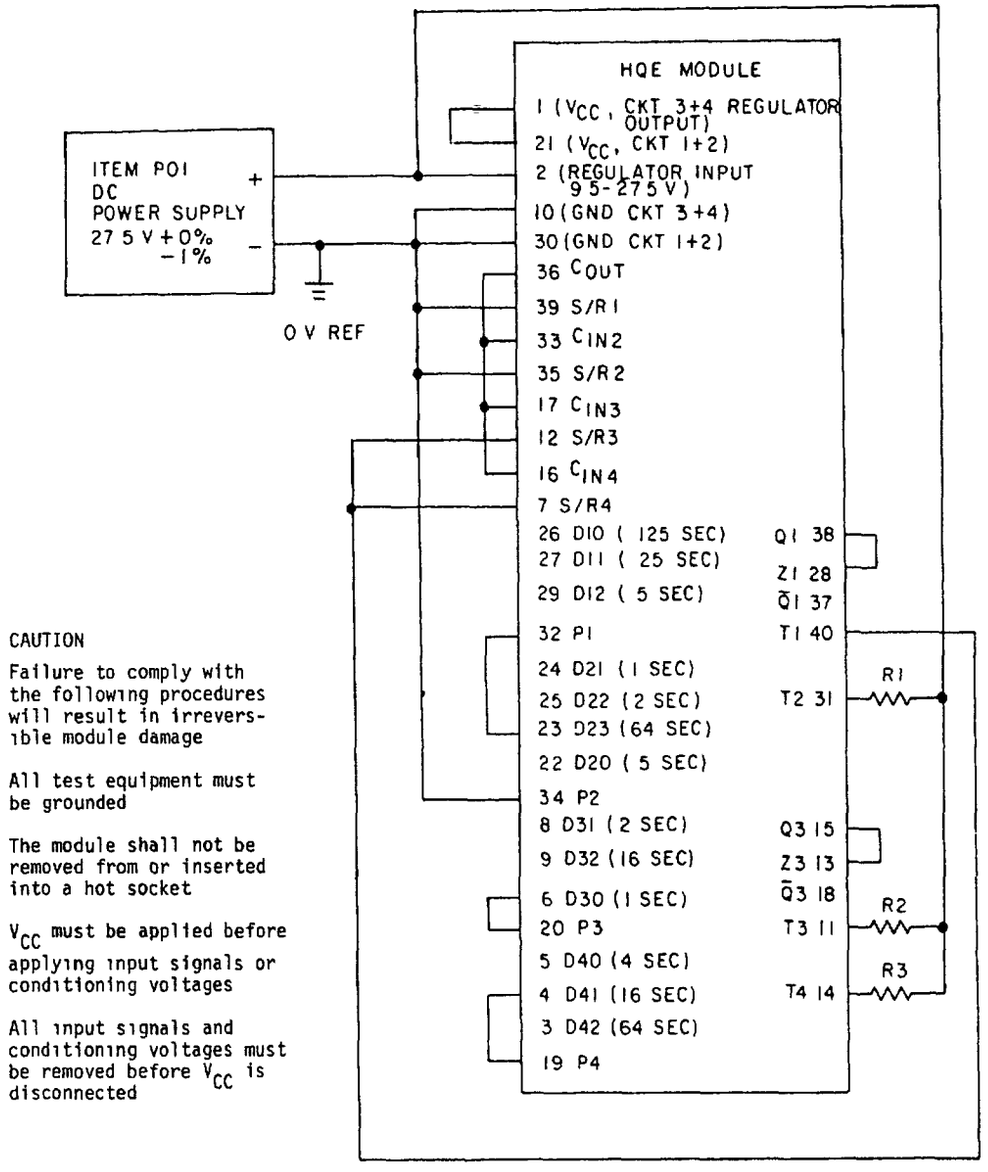



FIGURE 3 HOE module function diagram, programmable timer - Continued



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

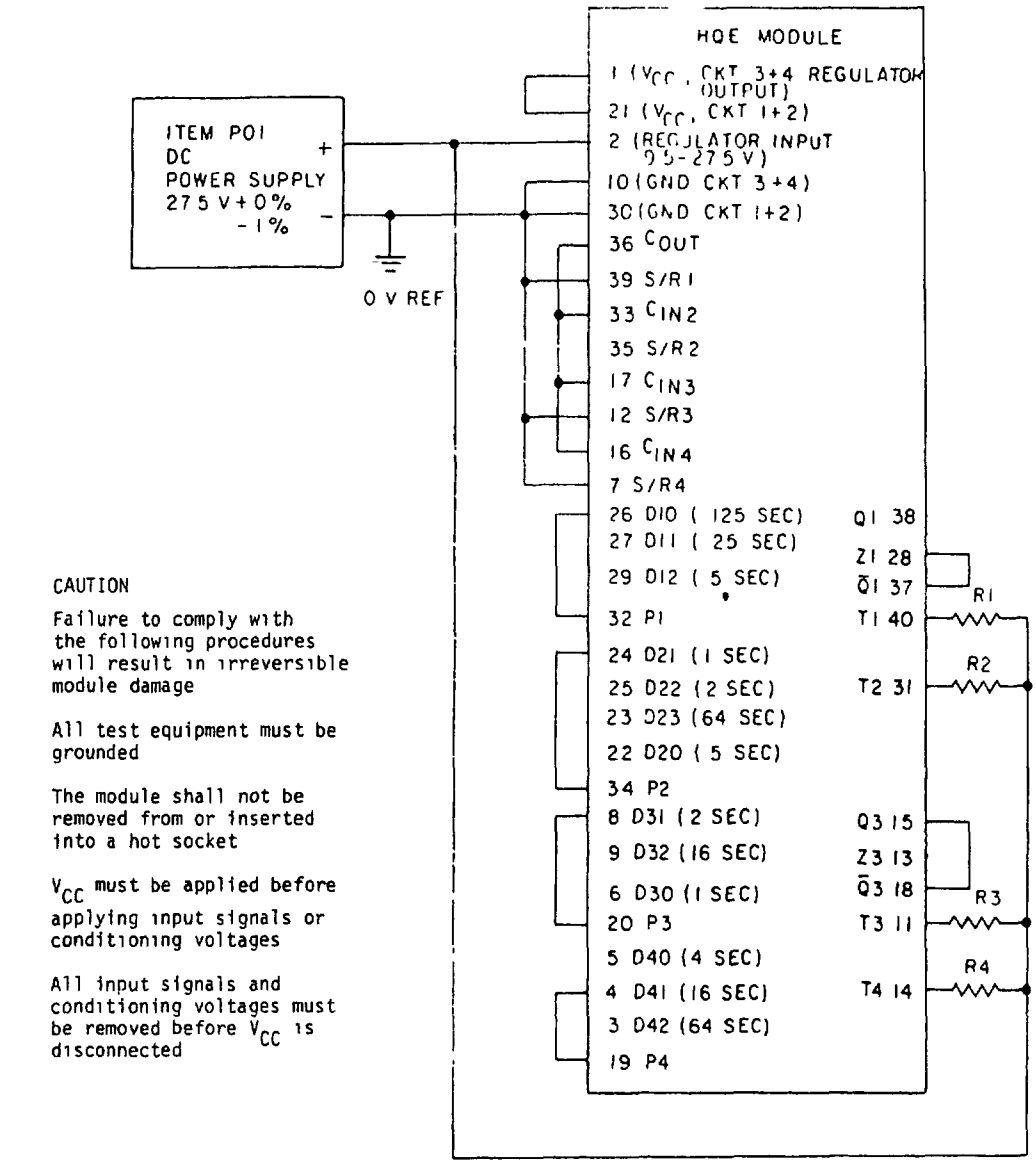



NOTE R1 - R3 = 1.1 kΩ ±1%, 1 W.

FIGURE 4. Operating temperature and life test circuit



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

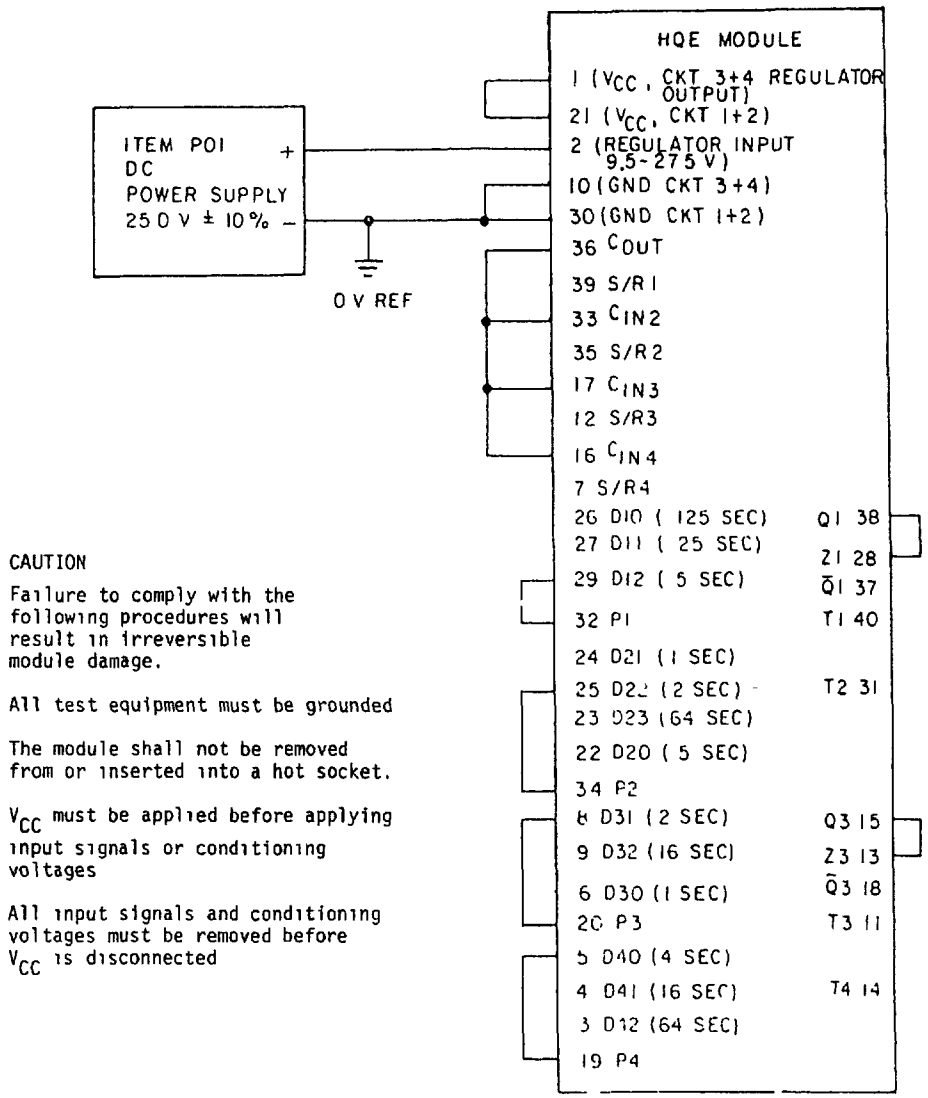



NOTE R1 - R4 = 1.1 kΩ ±1%, 1 W

FIGURE 5 Component temperature test circuit



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket.

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

FIGURE 6 Vibration test circuit



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

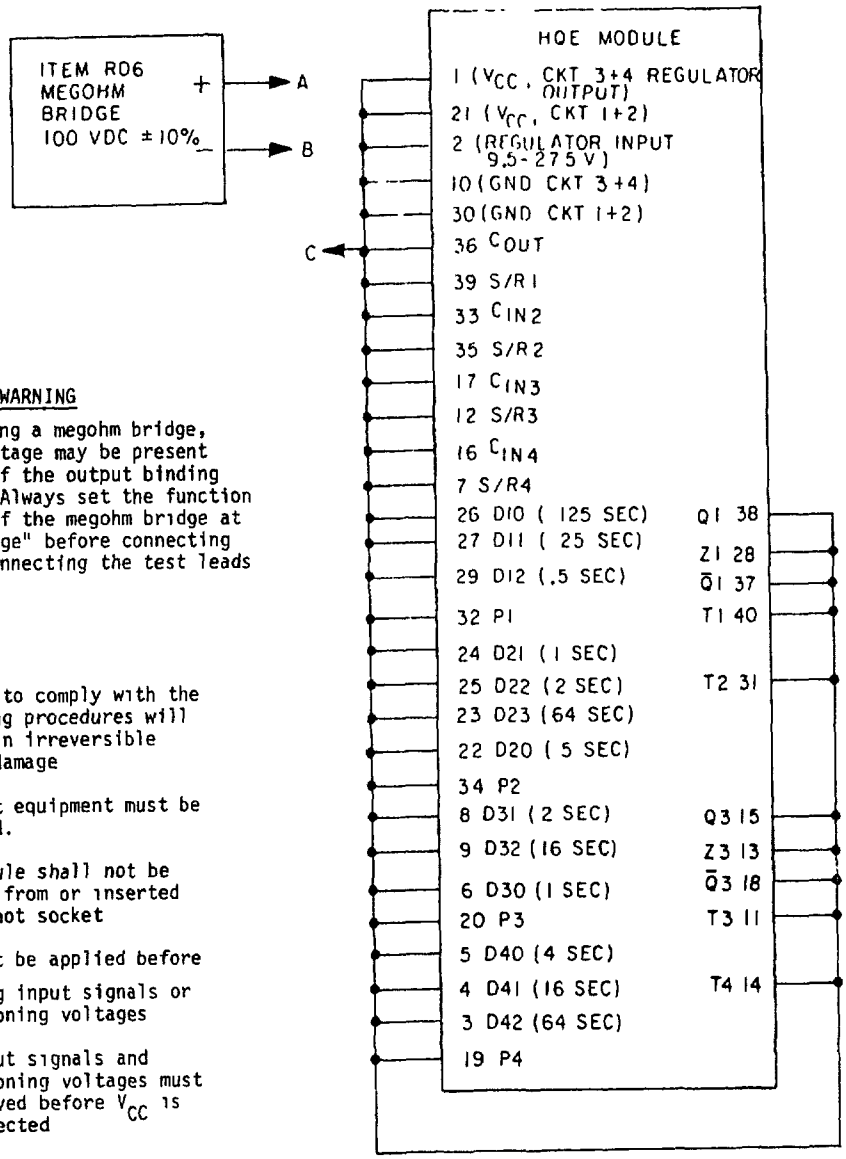
TABLE V Isolation test

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A | Isolation | C (all pins) | * Module fin | 10 | ∞ | MΩ | 7 |

* Penetrate the anodized finish on the module fin with a sharply pointed probe
Connect the negative lead of the megohm bridge to the probe



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads

CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

CAUTION

Make all pin connections to point C before making connection of point C to point A

FIGURE 7 Isolation test circuit for table V.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE VI 7 V $V_{CC}$ stress test

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | \ to pin | Min | Max | | |
| B | $V_{CC}$ stress | 36 | 7 0 | 9.0 | ms | 8 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

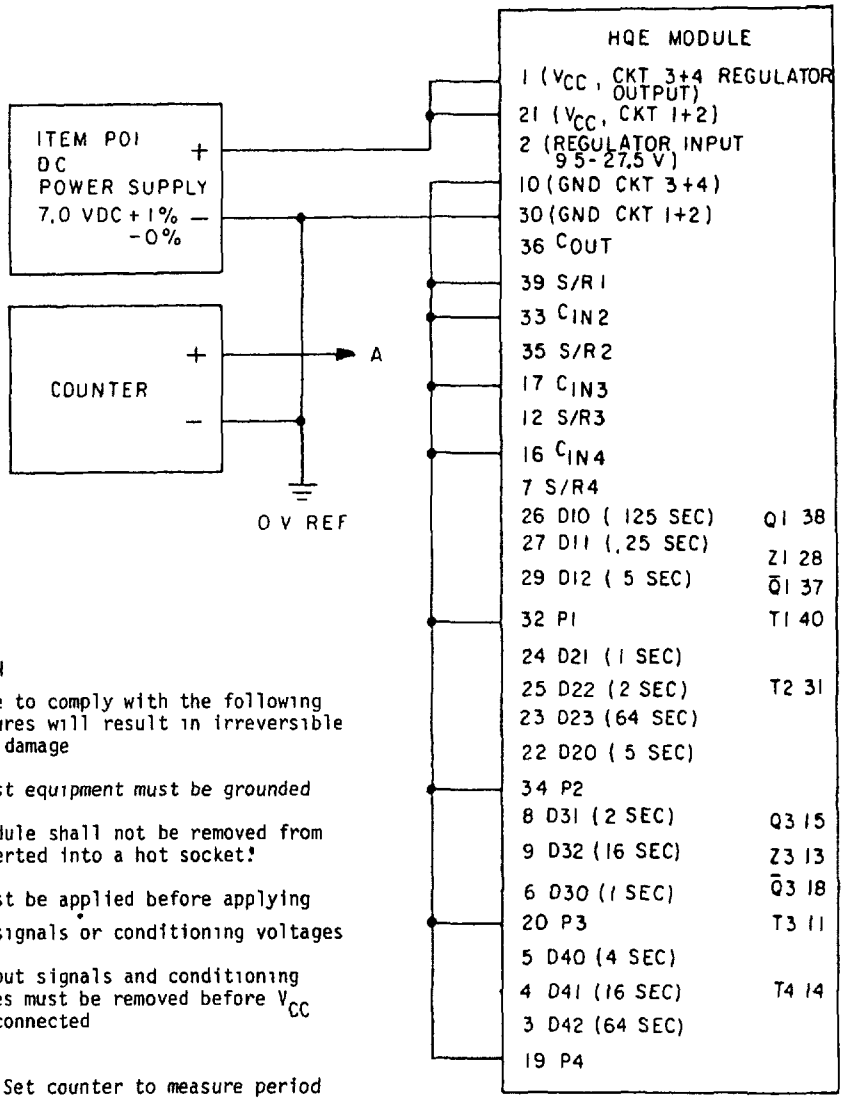



FIGURE 8 7 V $V_{CC}$ stress test circuit for table VI



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

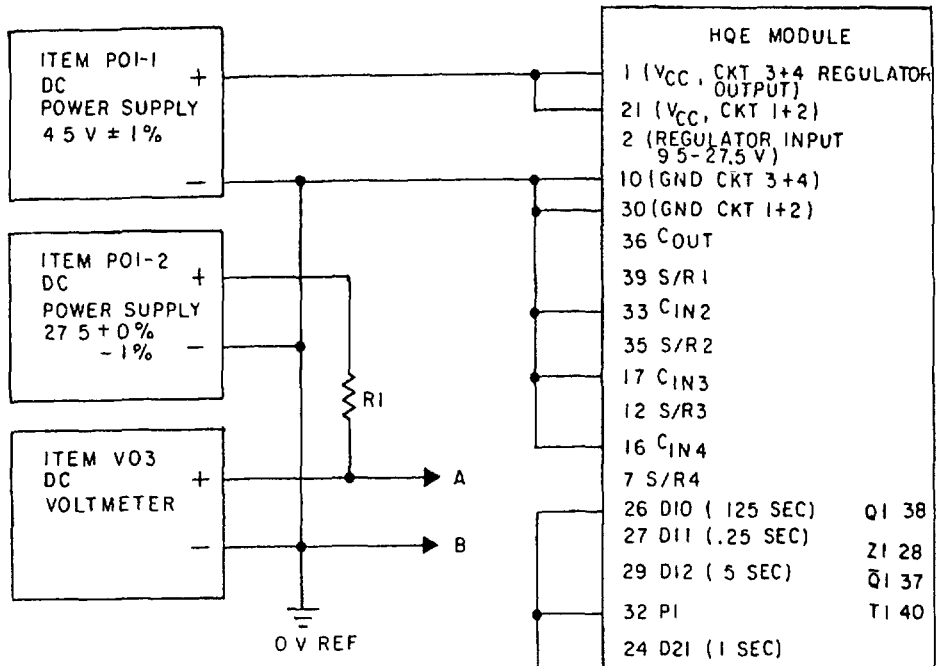
TABLE VII T output voltage, low level, $V_{CC}$ = 4 5 V test

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | B to pin | Min | Max | | |
| C1 | T1 $V_{OL}$ at $I_{OL}$ = 25 mA | 38-28 | 40 | Open | 0 0 | 0 4 | V | 9 |
| C2 | T2 | Open | 31 | Open | | | | |
| C3 | T3 | 15-13 | 11 | Open | | | | |
| C4 | T1 | 37-28 | 40 | 39 | | | | |
| C5 | T3 | 18-13 | 11 | 39 | | | | |
| C6 | T4 $V_{OL}$ at $I_{OL}$ = 25 mA | Open | 14 | 39 | 0 0 | 0 4 | V | 9 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTES
1. R1 = 1.1 kΩ ±1%, 1 W
2. Make pin to pin connections prior to connecting points A or B
3. For test conditions C4 - C6, make the output measurement a minimum of 0 135 seconds after pin 39 has been connected to point A

FIGURE 9 T output voltage low level, $V_{CC}$ = 4 5, test circuit for table VII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE VIII T, D, Q, and $\bar{Q}$ output voltage, low-level test

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| D1 | T2 output voltage, low level | 34 | Open | 31 | 19, 32, 20 | 0 0 | 0 4 | V | 10 |
| D2 | T4 | Open | 19 | 14 | 34, 32, 20 | | | | |
| D3 | T1 | | 28 | 40 | 19, 34, 32, 20 | | | | |
| D4 | T3 | | 13 | 11 | 19, 34, 20, 32 | | 0 4 | | |
| D5 | D10 | | 39 | 26 | | | 0 5 | | |
| D6 | D11 | | 39 | 27 | | | | | |
| D7 | D12 | | 39 | 29 | | | | | |
| D8 | D21 | | 35 | 24 | | | | | |
| D9 | D22 | | 35 | 25 | | | | | |
| D10 | D23 | | 35 | 23 | | | | | |
| D11 | D20 | | 35 | 22 | | | | | |
| D12 | D31 | | 12 | 8 | | | | | |
| D13 | D30 | | 12 | 6 | | | | | |
| D14 | D32 output voltage, low level | Open | 12 | 9 | 19, 34, 20, 32 | 0 0 | 0 5 | V | 10 |





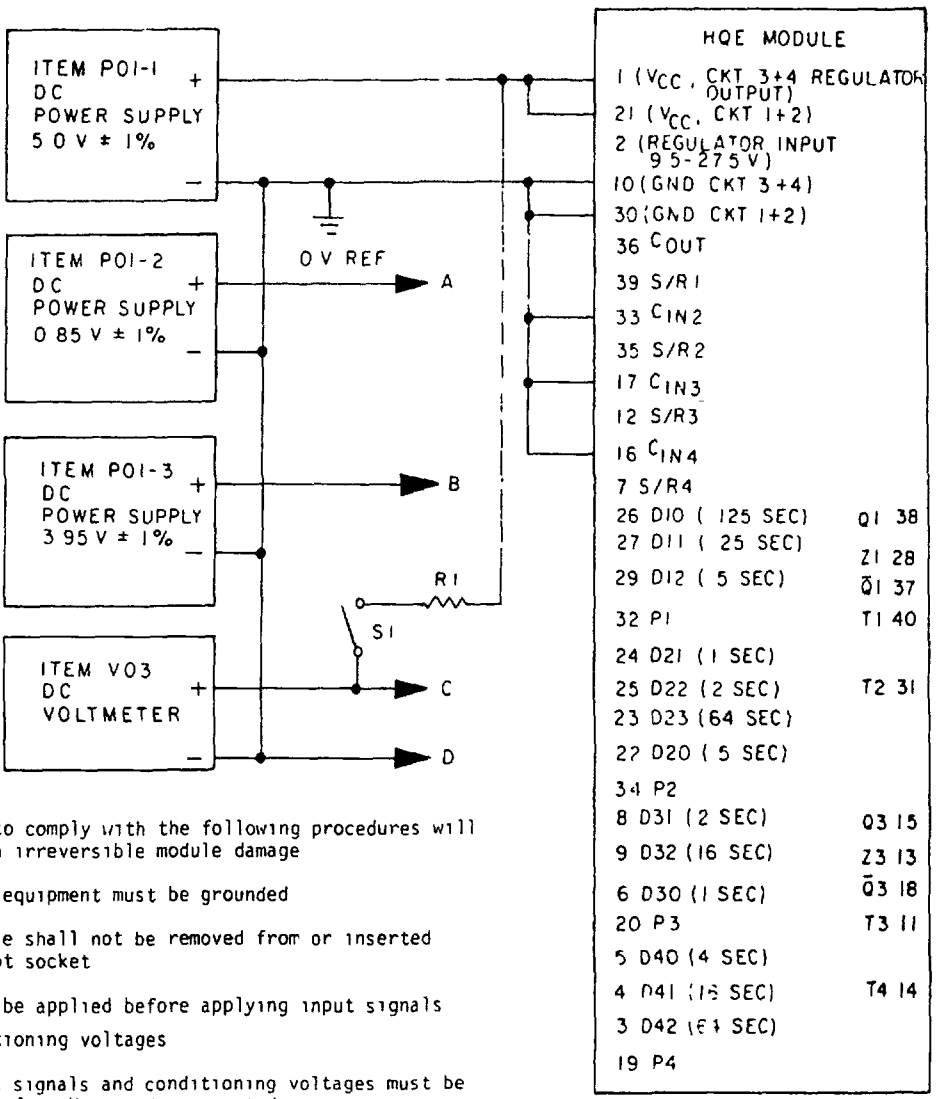
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE VIII T, D, Q, and $\bar{Q}$ output voltage, low-level test - Continued

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| D15 | D40 output voltage, low level | Open | 7 | 5 | 19, 34, 20, 32 | 0.0 | 0 5 | V | 10 |
| D16 | D41 | | 7 | 4 | 19, 34, 20, 32 | | 0 5 | | |
| D17 | D42 | | 7 | 3 | 19, 34, 20, 32 | | 0 5 | | |
| D18 | Q1 | Open | 32 | 38 | 20, 19, 34 | | 0 4 | | |
| D19 | $\bar{Q}$1 | 32 | Open | 37 | 20, 19, 34 | | | | |
| D20 | Q3 | Open | 20 | 15 | 32, 19, 34 | | | | |
| D21 | $\bar{Q}$3 | 20 | Open | 18 | 32, 19, 34 | | 0 4 | | |
| D22 | Q1 | Open | 32 | 38 | 20, 19, 34 | | 0 2 | | |
| D23 | $\bar{Q}$1 | 32 | Open | 37 | 20, 19, 34 | | | | |
| D24 | Q3 | Open | 20 | 15 | 32, 19, 34 | | | | |
| D25 | $\bar{Q}$3 output voltage, low level | 20 | Open | 18 | 32, 19, 34 | 0 0 | 0 2 | V | 10 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTES
1 For test conditions D1 - D4, R1 = 187 ±1%, 1/4 W, S1 closed
2 For test conditions D5 - D21, R1 value ±1%, 1/8 W listed below, S1 closed
3 For test conditions D22 - D25, S1 open
4 S1 = SPST

| TEST COND | 0° TO 25°C | 60° TO 80°C |
|---|---|---|
| D5 - D7 | 18 2 kΩ | 26 1 kΩ |
| D8 - D17 | 45 3 kΩ | 49 9 kΩ |
| D18 - D21 | 1 54 kΩ | 2.15 kΩ |

FIGURE 10 $V_{OUT}$, T, D, Q and $\bar{Q}$ voltage, low level, test circuit for table VIII



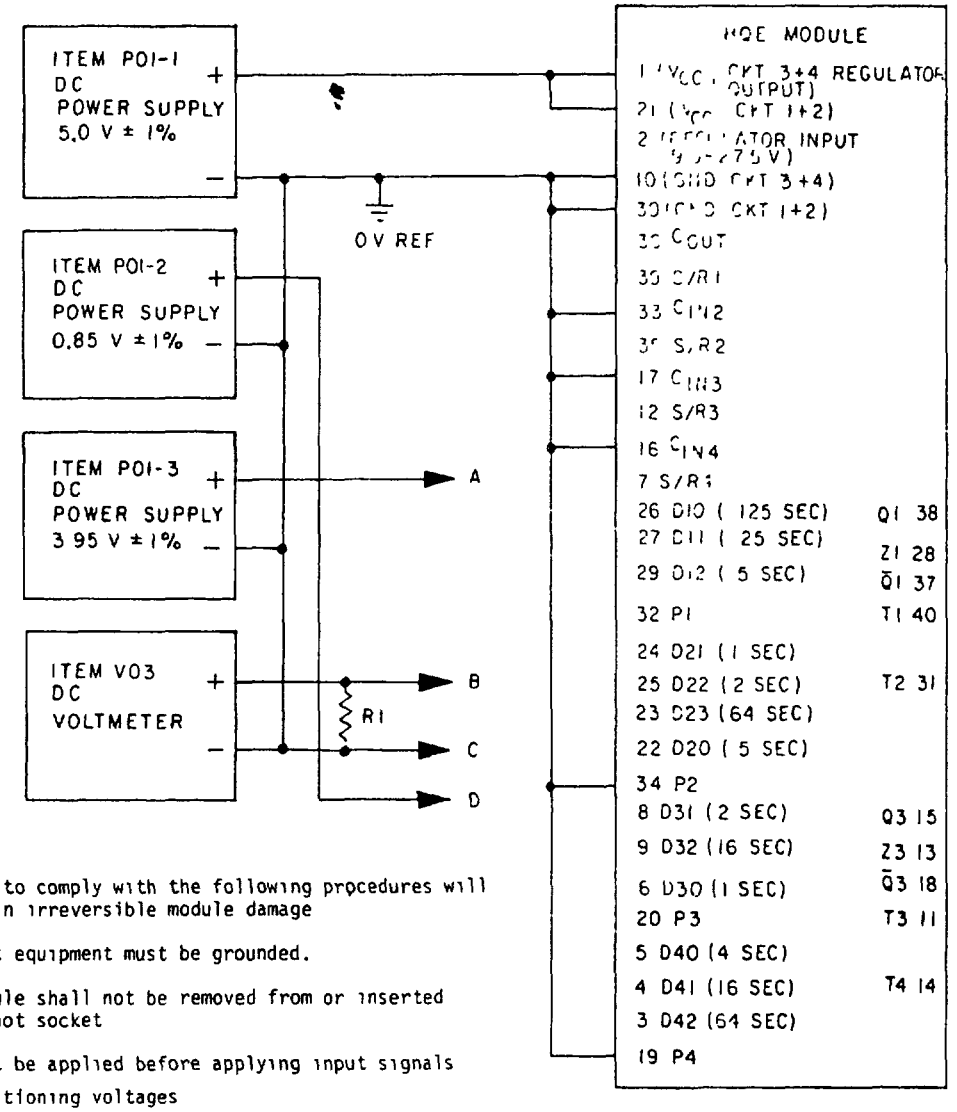
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE IX. Q, $\bar{Q}$, and D output voltage, high-level test

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| E1 | Q1 output voltage, high level | Open | Open | 38 | 20 | 32 | 4.2 | 5.0 | V | 11 |
| E2 | $\bar{Q}$ | | 32 | 37 | 20 | Open | | | | |
| E3 | Q3 | | Open | 15 | 32 | 20 | | | | |
| E4 | $\bar{Q}$4 | Open | 20 | 18 | 32 | Open | | | | |
| E5 | D10 | 26 to 32 | Open | 26 | 20 | 39 | | | | |
| E6 | D11 | 27 to 32 | Open | 27 | 20 | 39 | | | | |
| E7 | D12 output voltage, high level | 29 to 32 | Open | 29 | 20 | 39 | 4 2 | 5 0 | V | 11 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTES

1. For test conditions E1 - E7, R1 values ±1%, 1/8 W listed below

| TEST COND | 0° AND 25°C | 60° AND 80°C |
|---|---|---|
| E1 - E4 | 9.31 kΩ | 12.1 kΩ |
| E5 - E7 | 28.0 kΩ | 40.2 kΩ |

2 Test E5, make measurement 125 second after D is applied to pin 39
3 Test E6, make measurement 25 second after D is applied to pin 39
4 Test E7, make measurement 50 second after D is applied to pin 39

FIGURE 11 Q, $\overline{Q}$ and D output voltage, high level, test circuit for table IX



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE X D output voltage, high-level test

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | B to pin | C to pin | Min | Max | | |
| F1 | D23 output voltage, high level | 23 to 34 | 33 | 17, 20, 16, 19, 35 | 23 | 4 2 | 5 0 | V | 12, 13 |
| F2 | D21 | 24 to 34 | 33 | 17, 20, 16, 19 35 | 24 | | | | |
| F3 | D22 | 25 to 34 | 33 | 17, 20, 16, 19, 35 | 25 | | | | |
| F4 | D20 | 22 to 34 | 33 | 17, 20, 16, 19, 35 | 22 | | | | |
| F5 | D31 | 8 to 20 | 17 | 33, 34, 16, 19, 12 | 8 | | | | |
| F6 | D30 | 6 to 20 | 17 | 33, 34, 16, 19, 12 | 6 | | | | |
| F7 | D32 | 9 to 20 | 17 | 33, 34, 16, 19, 12 | 9 | | | | |
| F8 | D40 output voltage, high level | 5 to 19 | 16 | 33, 34, 7, 17, 20 | 5 | 4 2 | 5 0 | V | 12, 13 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

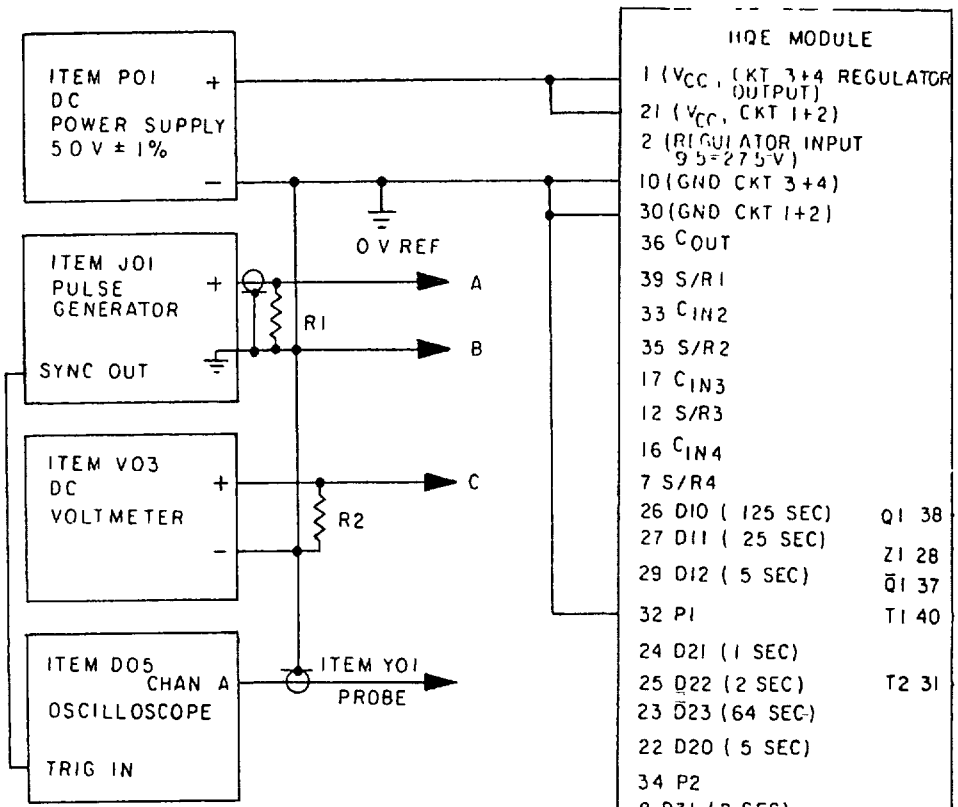
TABLE X D output voltage, high-level test.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | B to pin | C to pin | Min | Max | | |
| F9 | D41 output voltage, high level | 4 to 19 | 16 | 33, 34, 7, 17, 20 | 4 | 4 2 | 5 0 | V | 12, 13 |
| F10 | D42 output voltage, high level | 3 to 19 | 16 | 33, 34, 7, 17, 20 | 3 | 4 2 | 5 0 | V | 12, 13 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages.

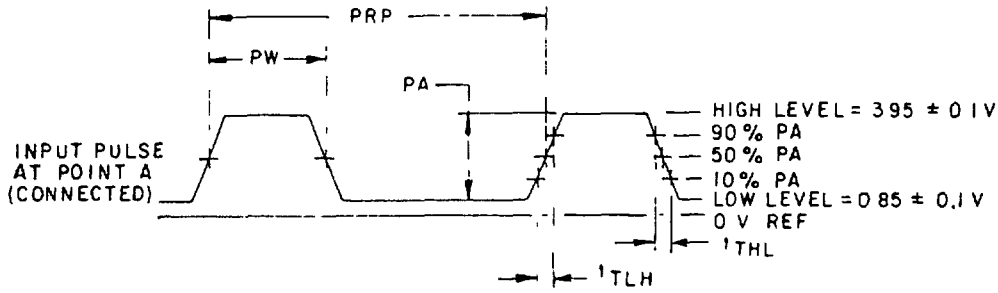
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected.

NOTES
1. R1 = 51Ω ±5%, 1/4 W, R2 at 0°-25°C = 46.4 kΩ ±1%, 1/8 W, R2 at 60°-80°C = 64.9 kΩ ±1%, 1/8 W.
2. Make all other circuit connections before connecting point B
3. Make measurement 3 µs after connecting point B

FIGURE 12 D output voltage, high level, test circuit for table X



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1 Input pulse parameters
PRP = 1 33 µs ±5%
PW = 0 66 µs ±5%
$t_{TLH}$, $t_{THL}$ = 15 ±2 ns
2 All voltages are referenced to 0 V unless otherwise specified

FIGURE 13 D output voltage, high level, waveforms for table X



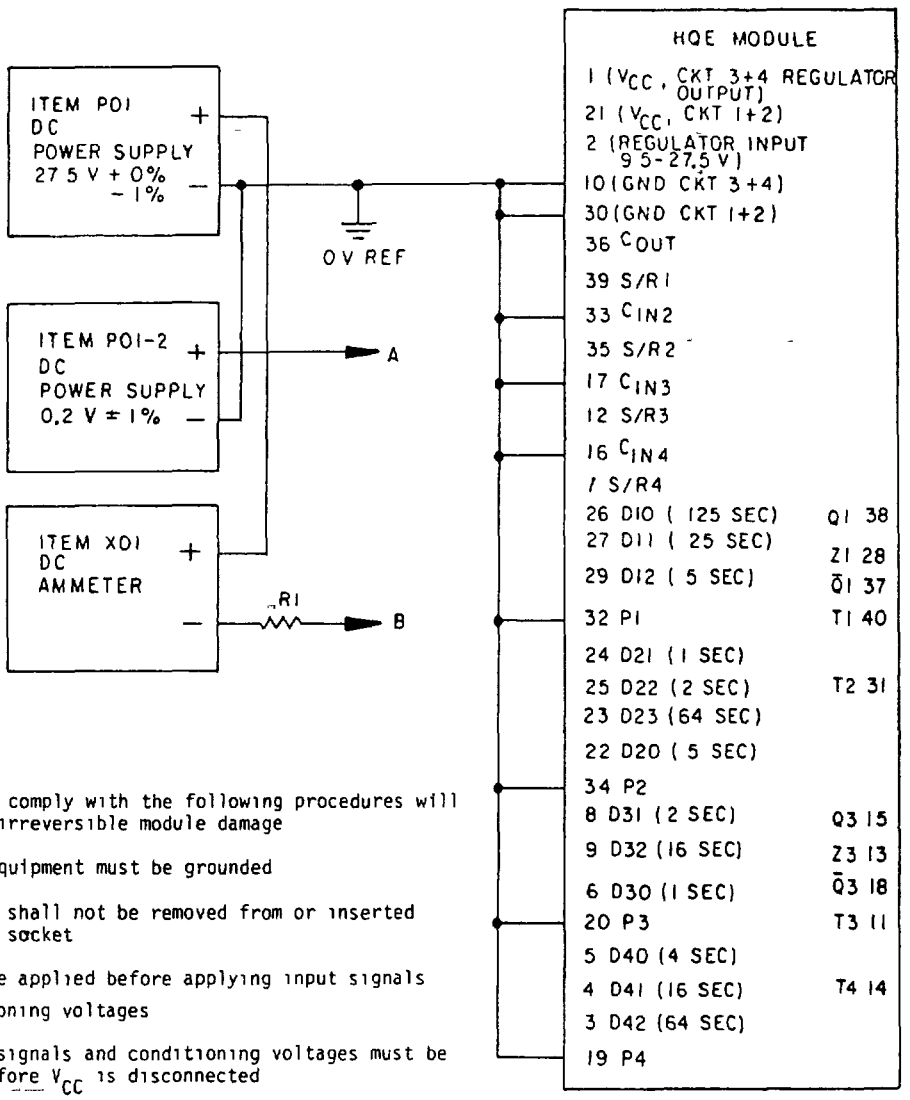
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XI Collector leakage current test

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| G1 | T1 collector leakage current | 28 | 40 | 0 | 25 | µA | 14 |
| G2 | T3 collector leakage current | 13 | 11 | 0 | 25 | µA | 14 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

CAUTION

Do not isolate chassis of ammeter X01 from earth ground

NOTE R1 = 10 kΩ ±1%, 1/4 W

FIGURE 14 Collector leakage current test circuit for table XI



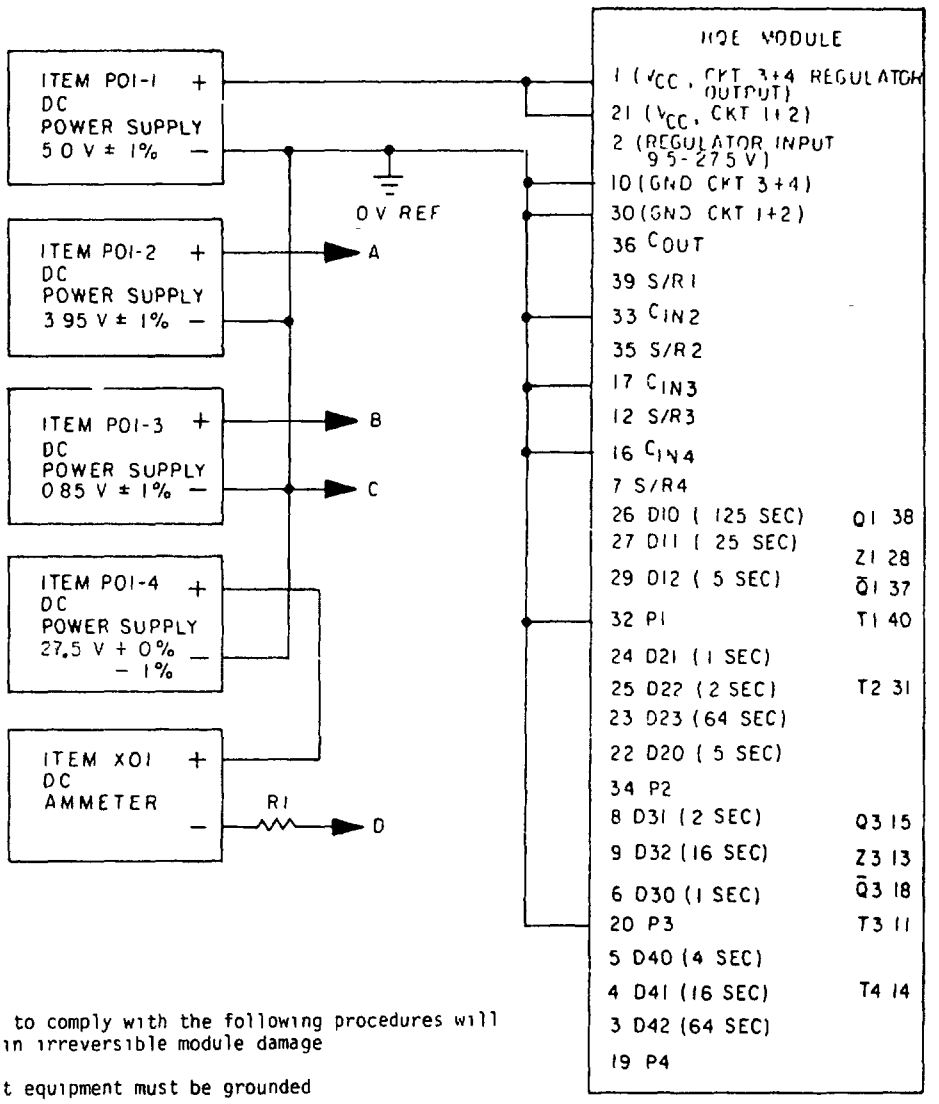
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XII. T collector leakage current test

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| H1 | T2 collector leakage current | 34 | Open | 19 | 31 | 0 | 25 | µA | 15 |
| H2 | T4 collector leakage current | Open | 19 | 34 | 14 | 0 | 25 | µA | 15 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

CAUTION

Do not isolate chassis of ammeter X01 from earth ground

NOTE R1 = 10 kΩ ±1%, 1/4 W

FIGURE 15 Collector leakage current test circuit for table XII



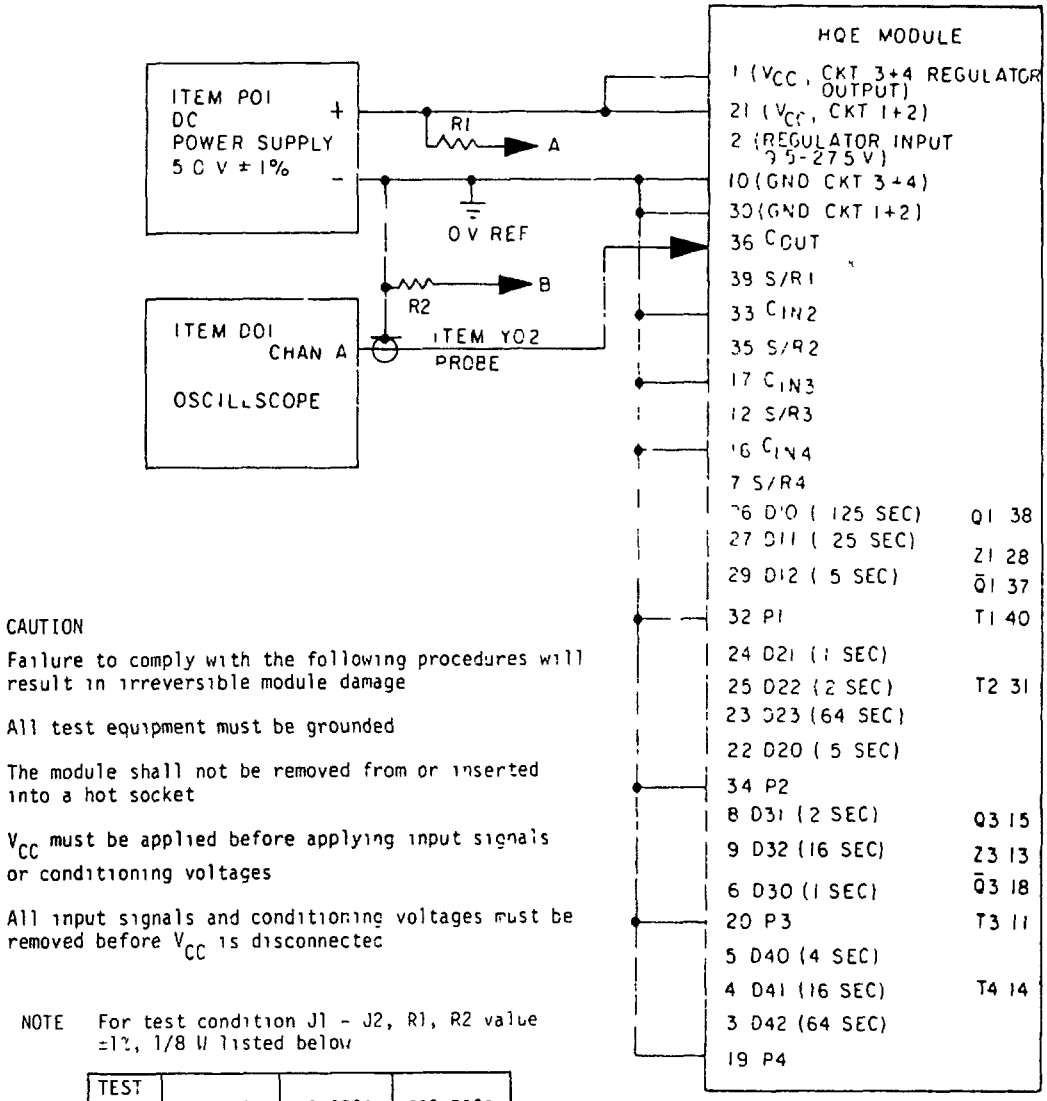
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XIII. $C_{OUT}$ (oscillator) voltage, low-and-high level test

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J1 | $C_{OUT}$ voltage, low level ($V_{OL}$) | 36 | Open | 0 0 | 0 4 | V | 16, 17 |
| J2 | $C_{OUT}$ voltage, high level ($V_{OH}$) | Open | 36 | 4.2 | 5 0 | V | 16, 17 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTE For test condition J1 - J2, R1, R2 value ±1%, 1/8 W listed below

| TEST COND | RESISTOR | 0°-25°C | 60°-80°C |
|---|---|---|---|
| J1 | R1 | 45.3 kΩ | 49.9 kΩ |
| J2 | R2 | 46.4 kΩ | 64.9 kΩ |

FIGURE 16 $C_{OUT}$ voltage low level and high level test circuit for table XIII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

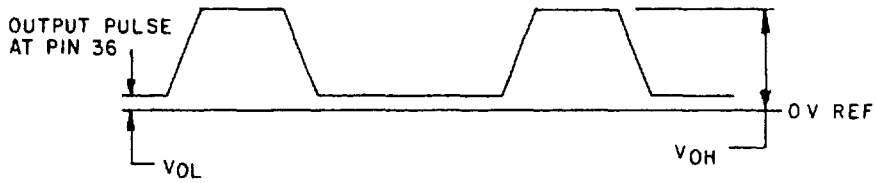



FIGURE 17. $C_{OUT}$ voltage low level and high level waveform for table XIII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

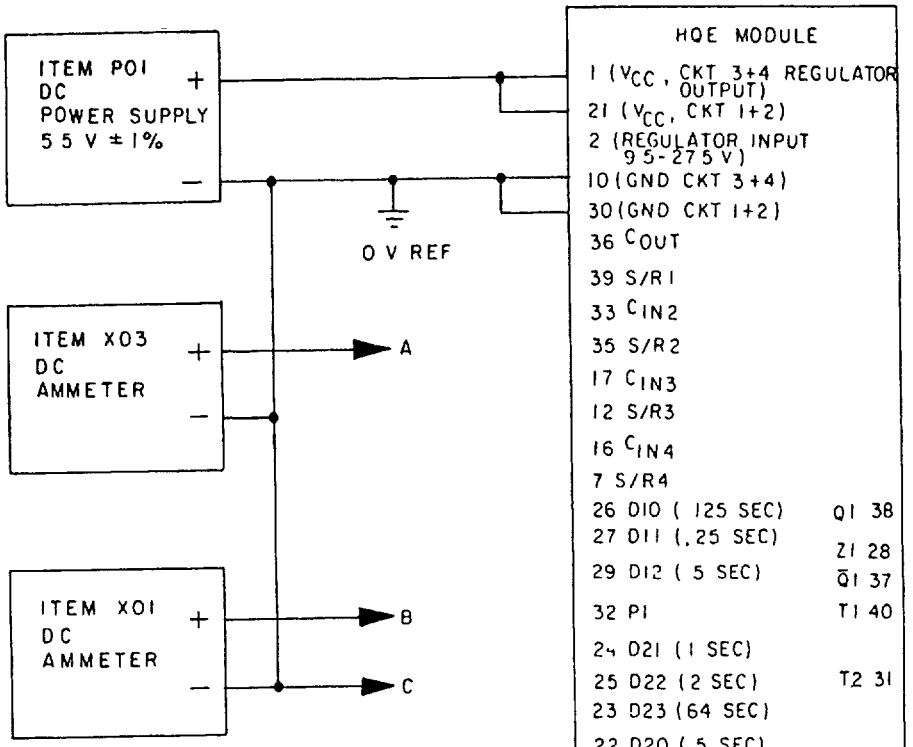
TABLE XIV P, $C_{IN}$, and S/R input current, low-level test

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| K1 | P1 input current, low level | 32 | Open | 34, 20, 19, 33, 17, 16 | 0 | 55 | nA | 18 |
| K2 | P2 | 34 | | 32, 20, 19, 33, 17, 16 | | | | |
| K3 | P3 | 20 | | 32, 34, 19, 33, 17, 16 | | | | |
| K4 | P4 | 19 | | 32, 34, 20, 33, 17, 16 | | | | |
| K5 | $C_{IN2}$ | 33 | | 32, 34, 20, 19, 17, 16 | | | | |
| K6 | $C_{IN3}$ | 17 | | 32, 34, 20, 19, 33, 16 | | | | |
| K7 | $C_{IN4}$ | 16 | Open | 32, 34, 20, 19, 33, 17 | 0 | 55 | nA | |
| K8 | S/R1 | Open | 39 | 33, 17, 16, 32, 34, 20, 19 | 1.25 | 1 40 | mA | |
| K9 | S/R2 | | 35 | | | | | |
| K10 | S/R3 | | 12 | | | | | |
| K11 | S/R4 input current, low level | Open | 7 | 33, 17, 16, 32, 34, 20, 19 | 1 25 | 1 40 | mA | 18 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

CAUTION

Do not isolate chassis of ammeters X01 and X03 from earth ground

FIGURE 18 P, $C_{IN}$ and S/R input current, low level, test circuit for table XIV



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

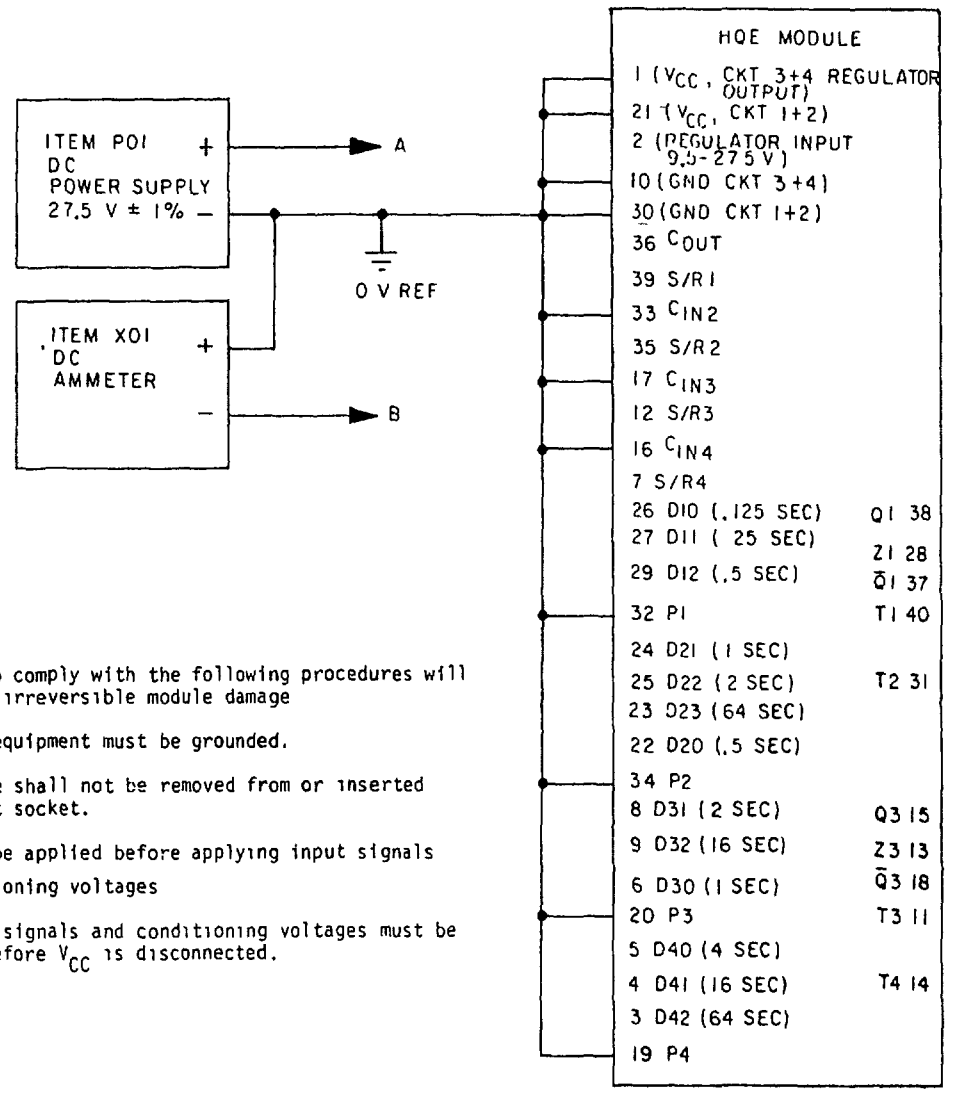
TABLE XV Z input current, low-level test.

| Test condition | Parameter | Connect | | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| L1 | Z1 input current, low level | 40 | 28 | 0 | 10 | µA | 19 |
| L2 | Z2 input current, low level | 11 | 13 | 0 | 10 | µA | 19 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected.

CAUTION

Do not isolate chassis of ammeter X01 from earth ground.

FIGURE 19 Z input current, low level, test circuit for table XV



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

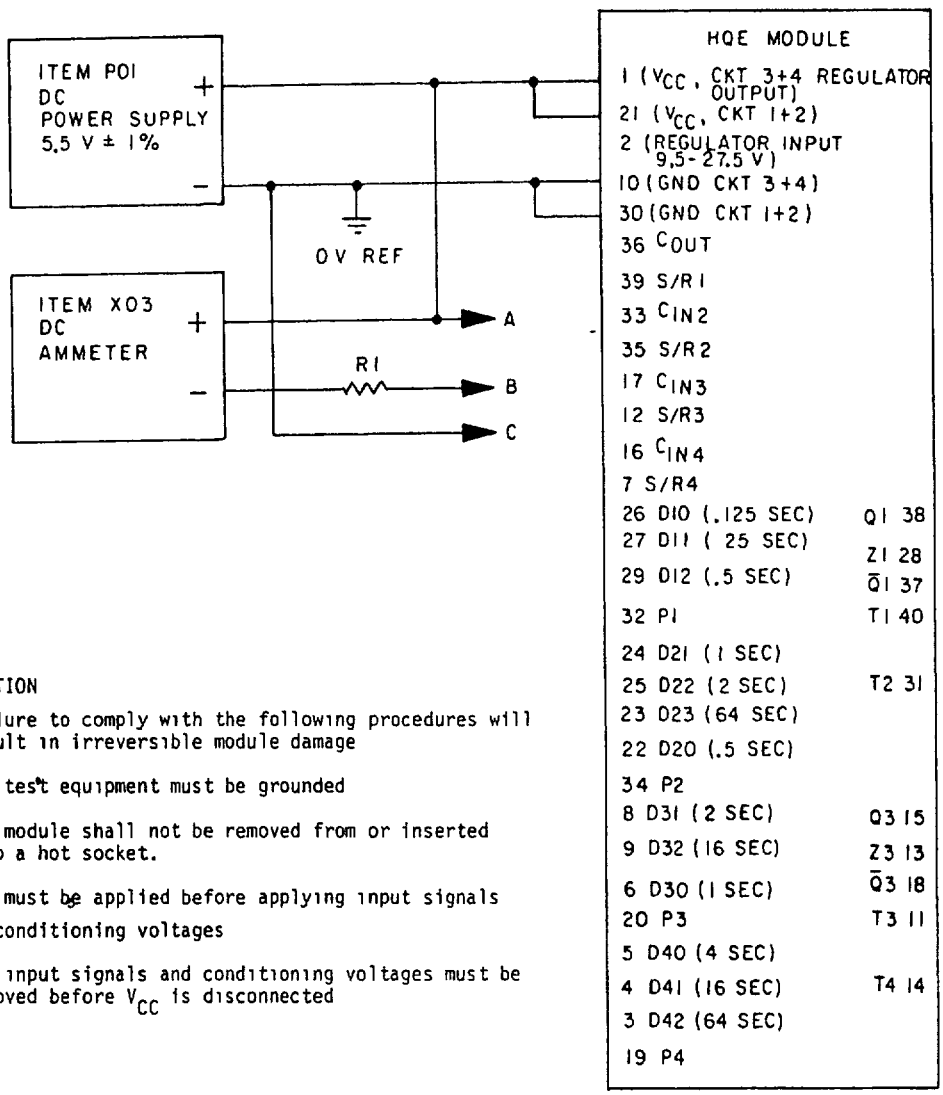
TABLE XVI. P, $C_{IN}$, and S/R input current, high-level test

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| M1 | P1 input current, high level | Open | 32 | 33,17,16,34,20,19 | 0 | 55 | nA | 20 |
| M2 | P2 | | 34 | 33,17,16,32,20,19 | | | | |
| M3 | P3 | | 20 | 33,17,16,32,34,19 | | | | |
| M4 | P4 | Open | 19 | 33,17,16,32,34,20 | | | | |
| M5 | $C_{IN2}$ | 34 | 33 | 32,17,20,16,19 | | | | |
| M6 | $C_{IN3}$ | 20 | 17 | 32,33,34,16,19 | | | | |
| M7 | $C_{IN4}$ | 19 | 16 | 32,33,34,17,20 | | 55 | | |
| M8 | S/R1 | Open | 39 | 33,17,16,32,34,20,19 | | 200 | | |
| M9 | S/R2 | | 35 | | | | | |
| M10 | S/R3 | | 12 | | | | | |
| M11 | S/R4 input current, high level | Open | 7 | 33,17,16,32,34,20,19 | 0 | 200 | nA | 20 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket.

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

CAUTION.

Do not isolate chassis of ammeter X03 from earth ground.

NOTE R1 = 10 kΩ ±1%, 1/4 W.

FIGURE 20 P, $C_{IN}$, S/R input current, high level, test circuit for table XVI



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

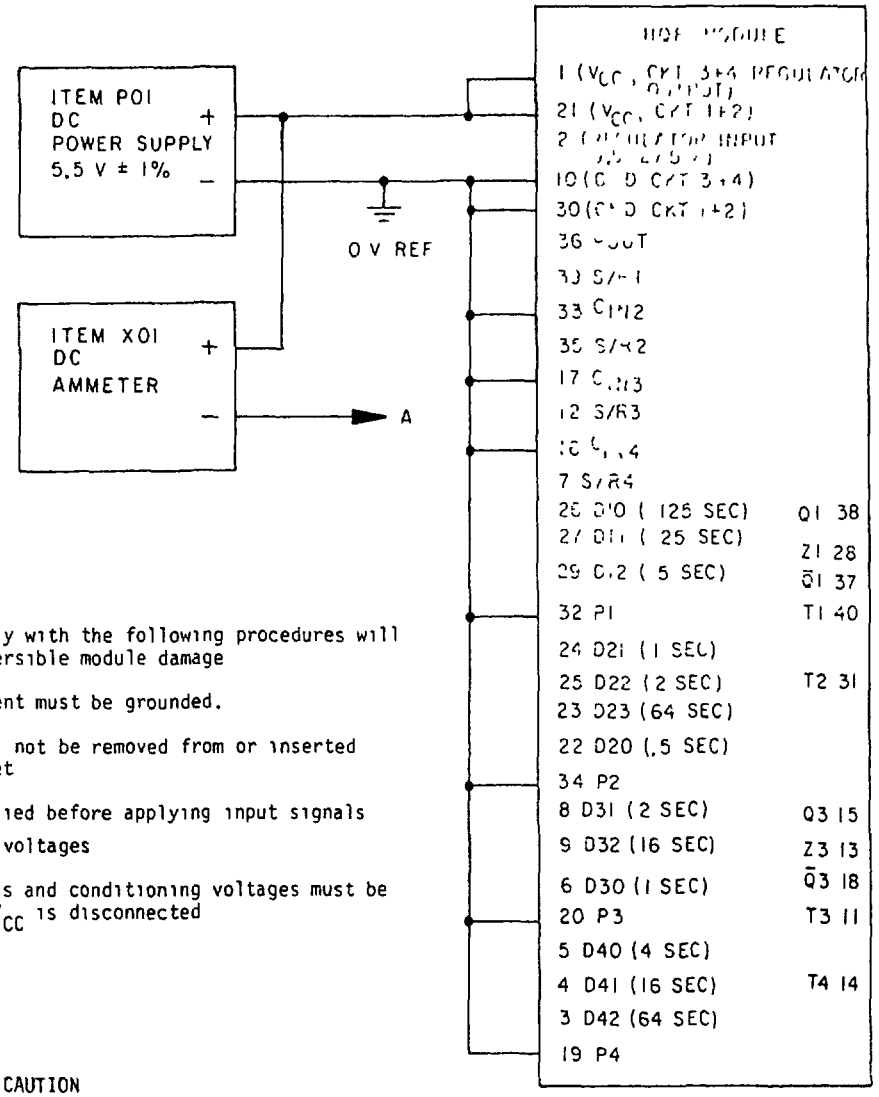
TABLE XVII. Z input current, high-level test.

| Test condition | Parameter | Connect A to pin | Initial limits, 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | | Min | Max | | |
| N1 | Z1 input current, high level | 28 | 1 0 | 1 4 | mA | 21 |
| N2 | Z2 input current, high level | 13 | 1 0 | 1 4 | mA | 21 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

CAUTION

Do not isolate chassis of ammeter X01 from earth ground

FIGURE 21 Z input current, high level, test circuit for table XVII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

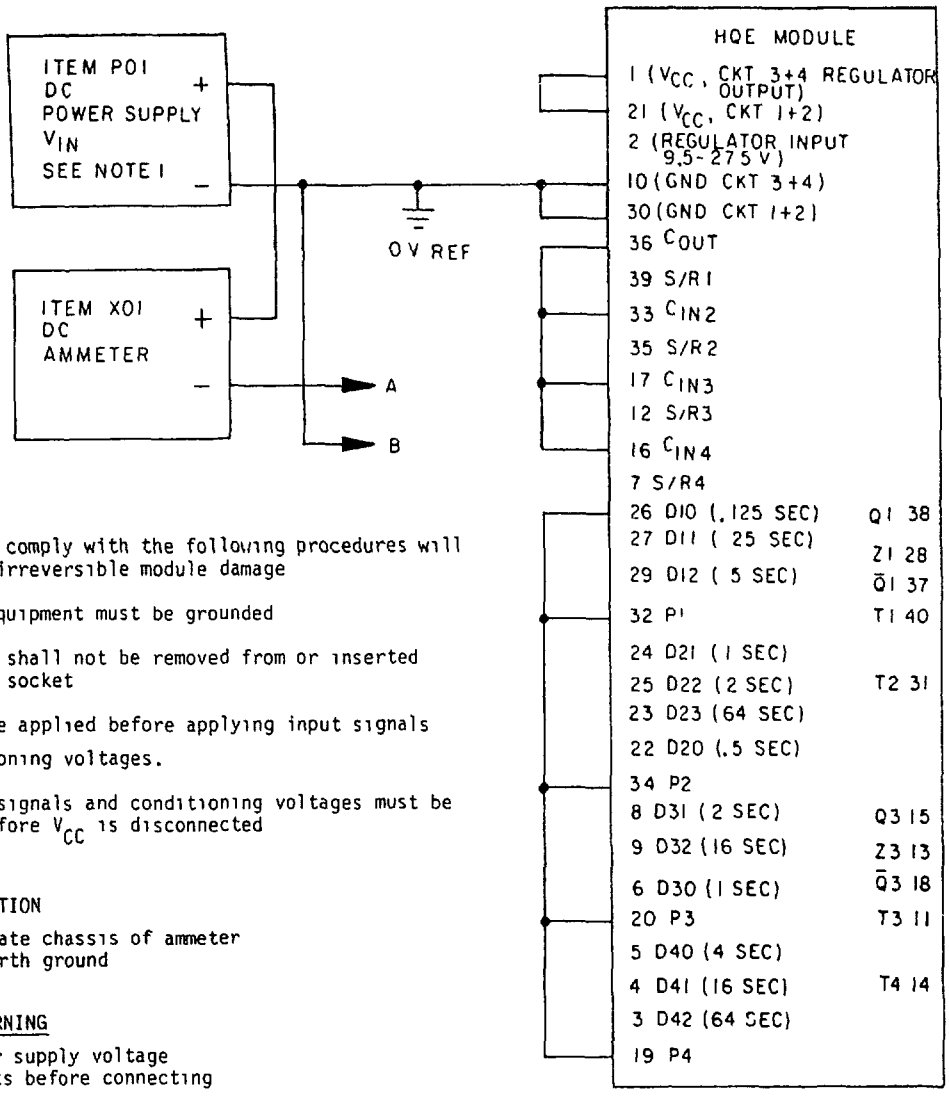
TABLE XVIII Power supply current test.

| Test condition | Parameter | Connect | | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| P1 | Power supply current test $I_{CC}$, $V_{CC}$ = 27 5 V, reset mode | 2 | Open | 5 | 20 | mA | 22 |
| P2 | Power supply current test $I_{CC}$, $V_{CC}$ = 27 5, set mode | 2 | 39, 35, 12, 7 | 7 | 25 | | |
| P3 | Power supply current test $I_{CC}$, $V_{CC}$ = 5 5 V reset mode | 1 | Open | 5 | 19 | | |
| P4 | Power supply current test $I_{CC}$, $V_{CC}$ = 5 5 V, set mode | 1 | 39, 35, 12, 7 | 7 | 26 | mA | 22 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages.

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

CAUTION

Do not isolate chassis of ammeter X01 from earth ground

WARNING

Make power supply voltage adjustments before connecting point A.

NOTES

1 For test conditions P1 and P2 $V_{IN}$ = 27.5 V +0, -1%

2 For test conditions P3 and P4 $V_{IN}$ = 5.5 V ±1%.

3 Make measurement after point B connection is made for a minimum of 0.13 second

FIGURE 22 Power supply current test circuit for table XVIII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

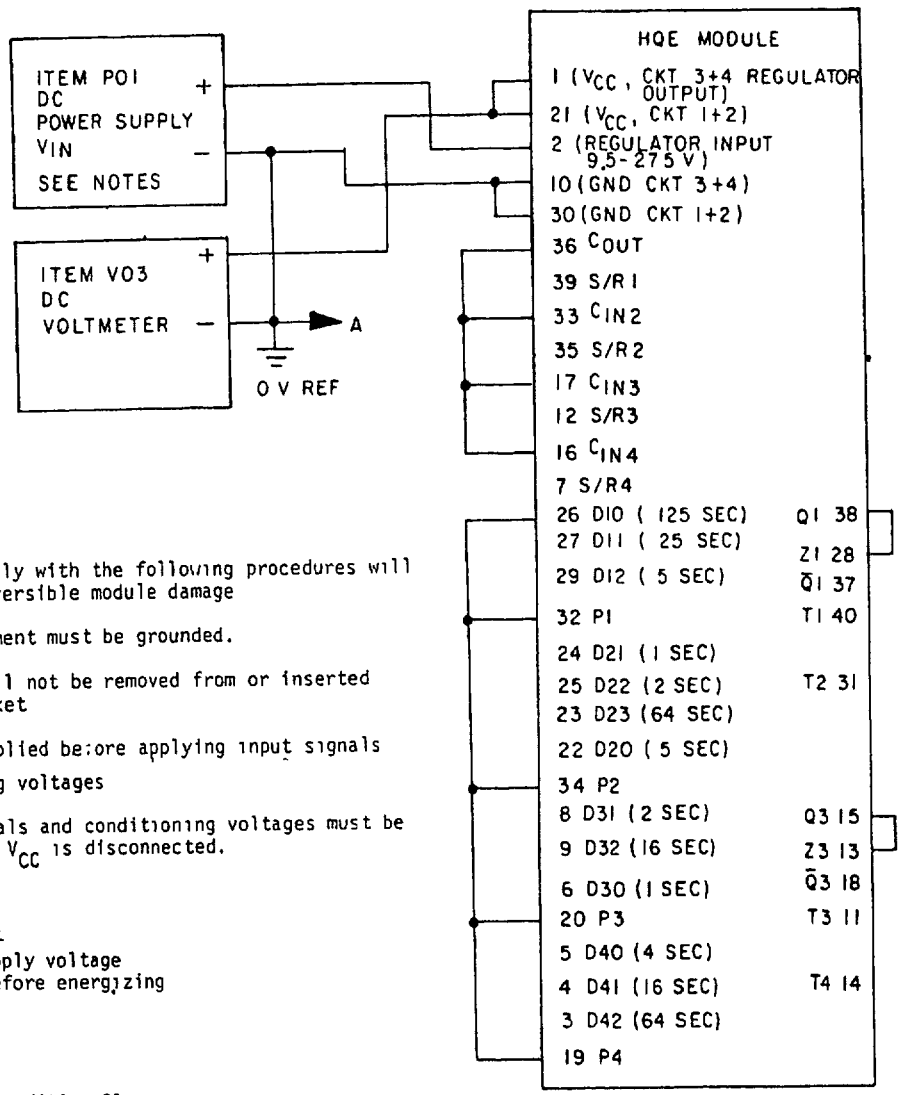
TABLE XIX Power supply (low-level and high-level) regulation test

| Test condition | Parameter | Connect | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | X to pin | Min | Max | | |
| Q1 | Power supply regulation, low level mode | Open | 4.5 | 5 5 | V | 23 |
| Q2 | Power supply regulation, high level mode | 12,7,39,35 | 4 5 | 5 5 | V | 23 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION:

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected.

WARNING

Make power supply voltage adjustments before energizing test circuit.

NOTES

1. For test condition Q1, $V_{IN}$ = 9.5 V, +1%, -0
2. For test condition Q2, $V_{IN}$ = 27.5 V, +0, -1%
3. For test condition Q2, measure output 0.13 second after pin A is connected

FIGURE 23 Power supply, low level and high level, regulation test circuit for table XIX



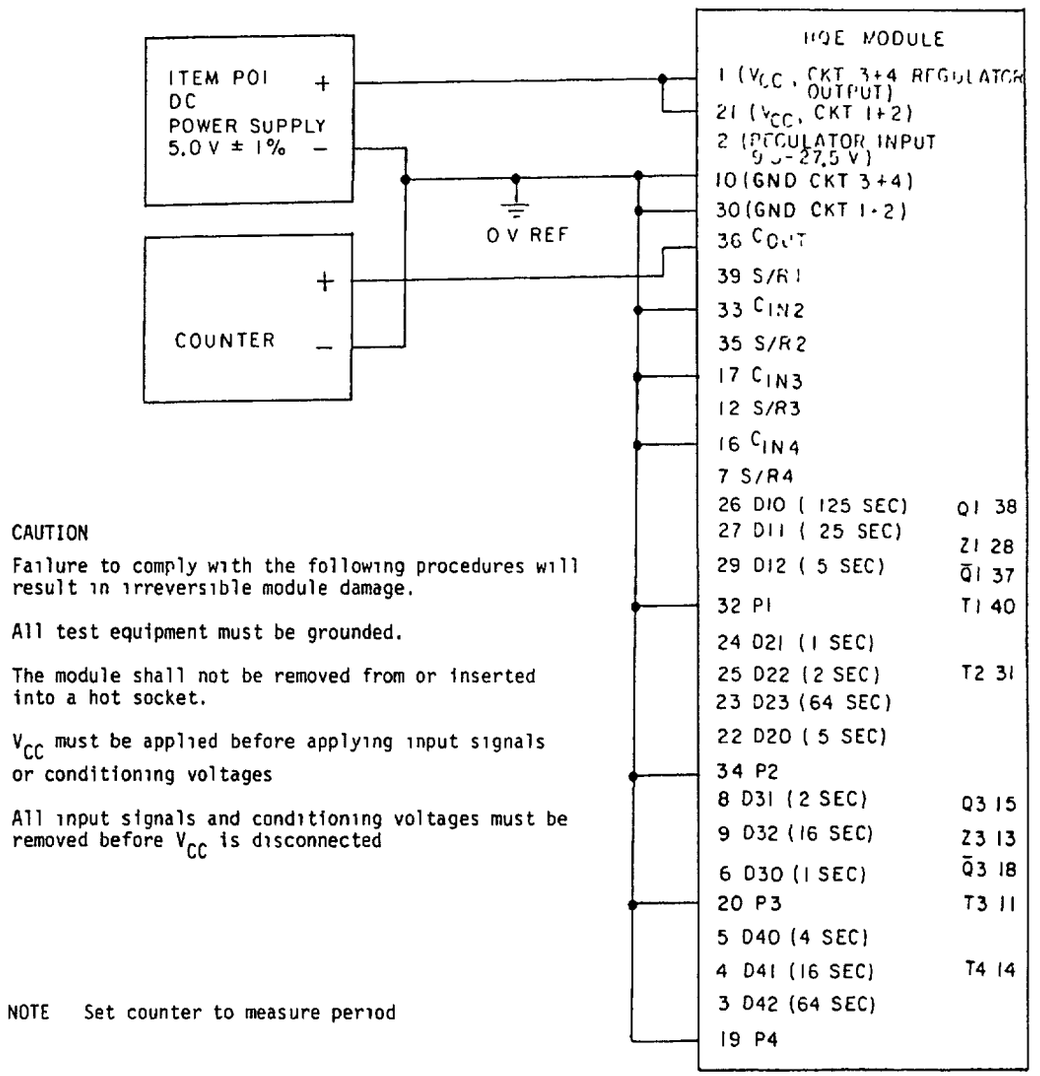
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XX. Oscillator repetition period test

| Test condition | Parameter | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| R | Oscillator repetition period | 7 81 | 7 82 | ms | 24 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

$V_{CC}$ must be applied before applying input signals or conditioning voltages

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTE Set counter to measure period

FIGURE 24 Oscillator repetition period test circuit for table XX



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

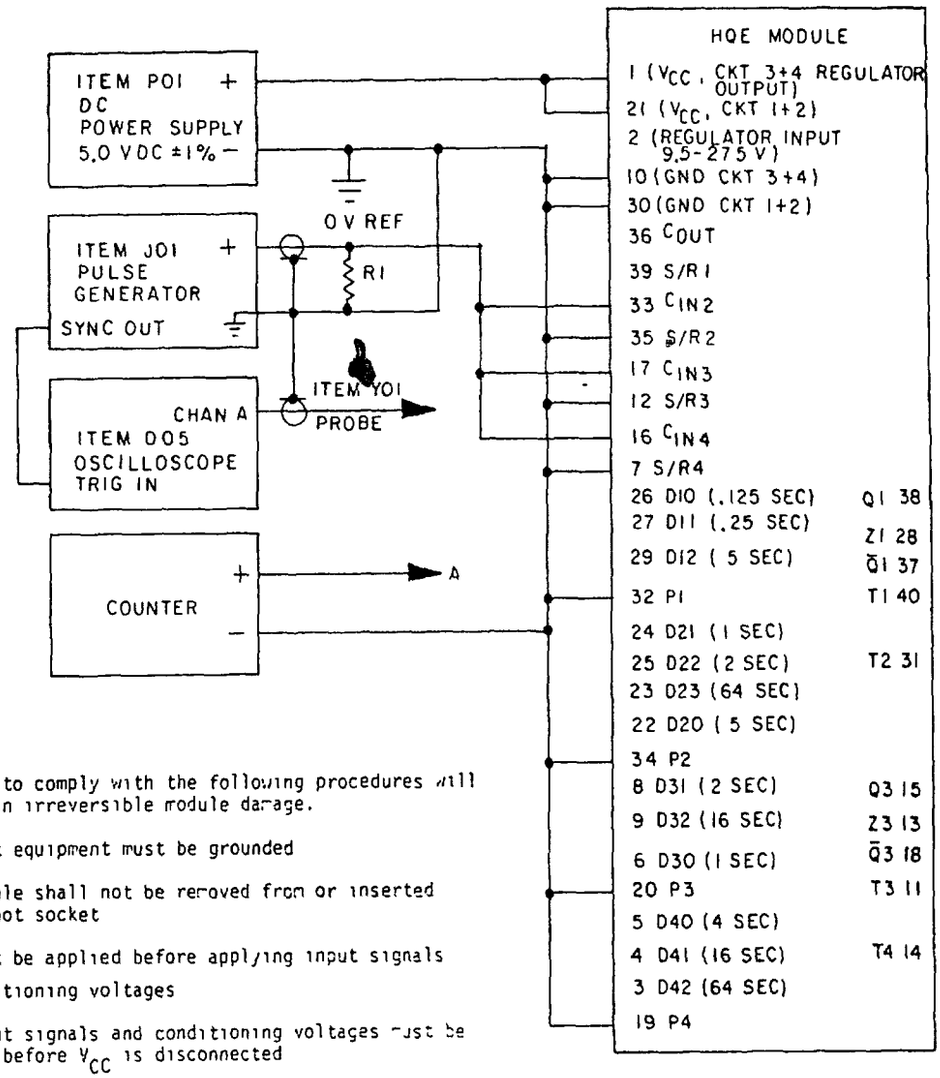
TABLE XXI D output repetition period test

| Test condition | Parameter | Connect | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| S1 | D20 output repetition period | 22 | 162 | 179 | µs | 25, 26 |
| S2 | D21 | 24 | 324 | 359 | µs | |
| S3 | D22 | 25 | 649 | 718 | µs | |
| S4 | D23 | 23 | 20.79 | 22 98 | ms | |
| S5 | D31 | 8 | 649 | 718 | µs | |
| S6 | D30 | 6 | 5 19 | 5 79 | ms | |
| S7 | D32 | 9 | 324 | 359 | µs | |
| S8 | D40 | 5 | 1.29 | 1 44 | ms | |
| S9 | D1 | 4 | 5 19 | 5 79 | ms | |
| S10 | D42 output repetition period | 3 | 20.79 | 22 98 | ms | 25, 26 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

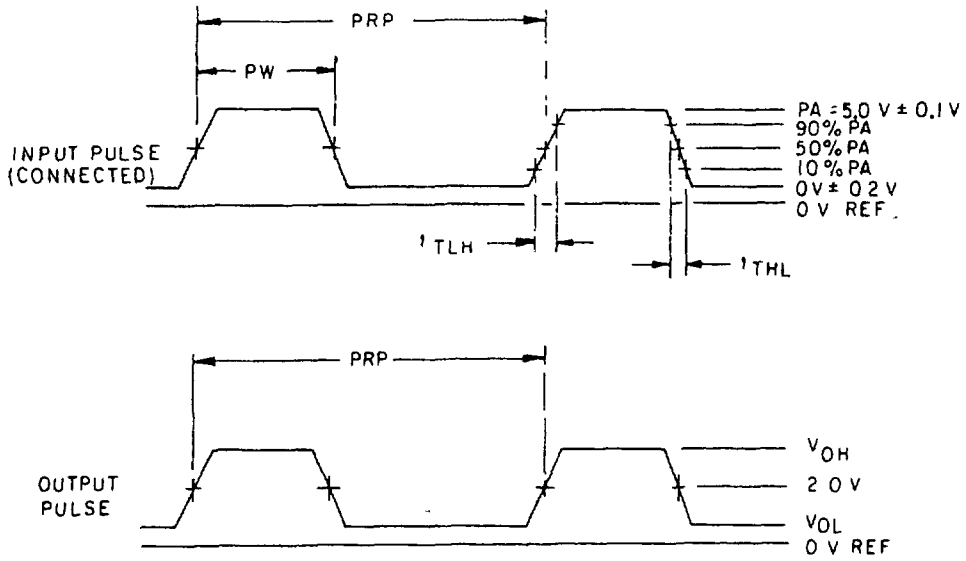
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTES
1. R1 = 51Ω ±5%, 1/4 W
2 Set counter to measure period

FIGURE 25 D output repetition period test circuit for table XXI



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTE Input pulse parameter
PRP = 1.33 us ±5%
PW = 0.66 us ±5%
$t_{TLH}$, $t_{THL}$ = 15 ±2 ns

FIGURE 26 Output repetition period waveforms for table XXI.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

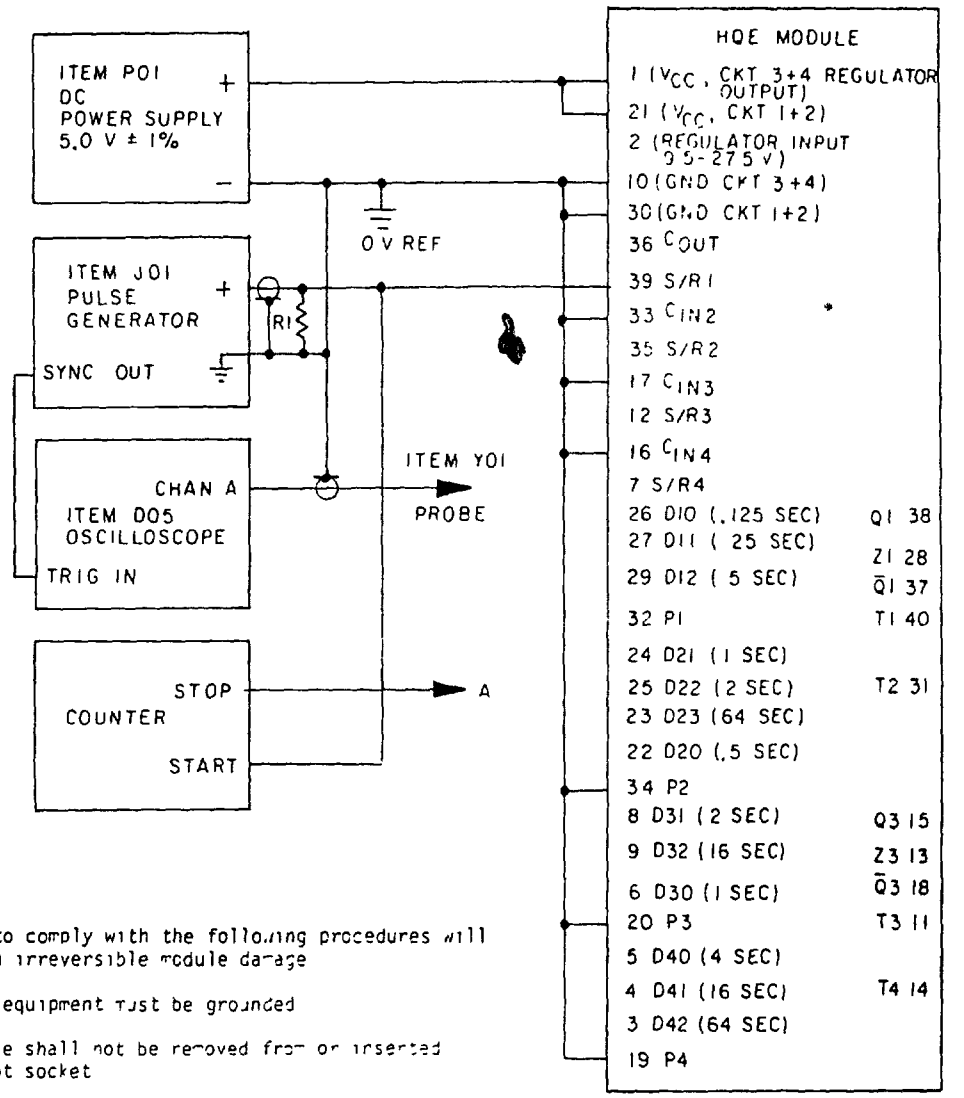
TABLE XXII. D output time interval and minimum S/R pulse width test

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | Min | Max | | |
| T1 | D10 output time interval | 26 to 32 | 26 | 115 | 135 | ms | 27, 28 |
| T2 | D12 output time interval | 27 to 32 | 27 | 240 | 260 | ms | 27, 28 |
| T3 | D13 output time interval | 29 to 32 | 29 | 490 | 510 | ms | 27, 28 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

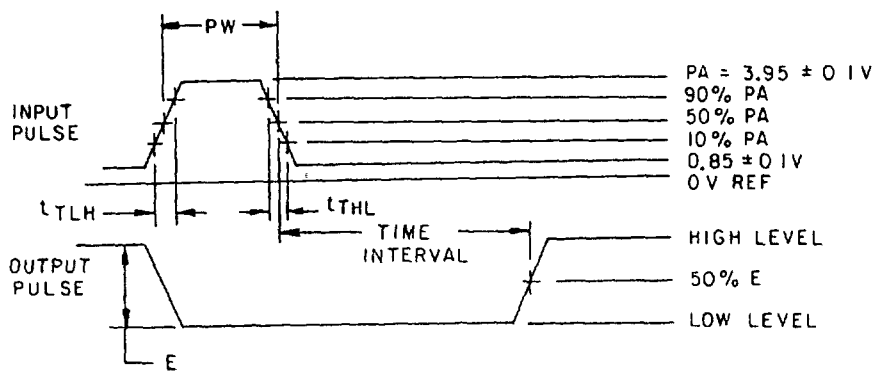
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTES
1 R1 = 51Ω ±5%, 1/4 W
2 Set counter to measure time interval

FIGURE 27 D output time interval and minimum S/R pulse with test circuit for table XXII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTE Input pulse parameters
Set item J01 for single pulse mode
PW = 2.8 +.2, -0 µs
$t_{TLH}$, $t_{THL}$ = 15 ±2 ns

FIGURE 28. D output time interval waveforms for table XXII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

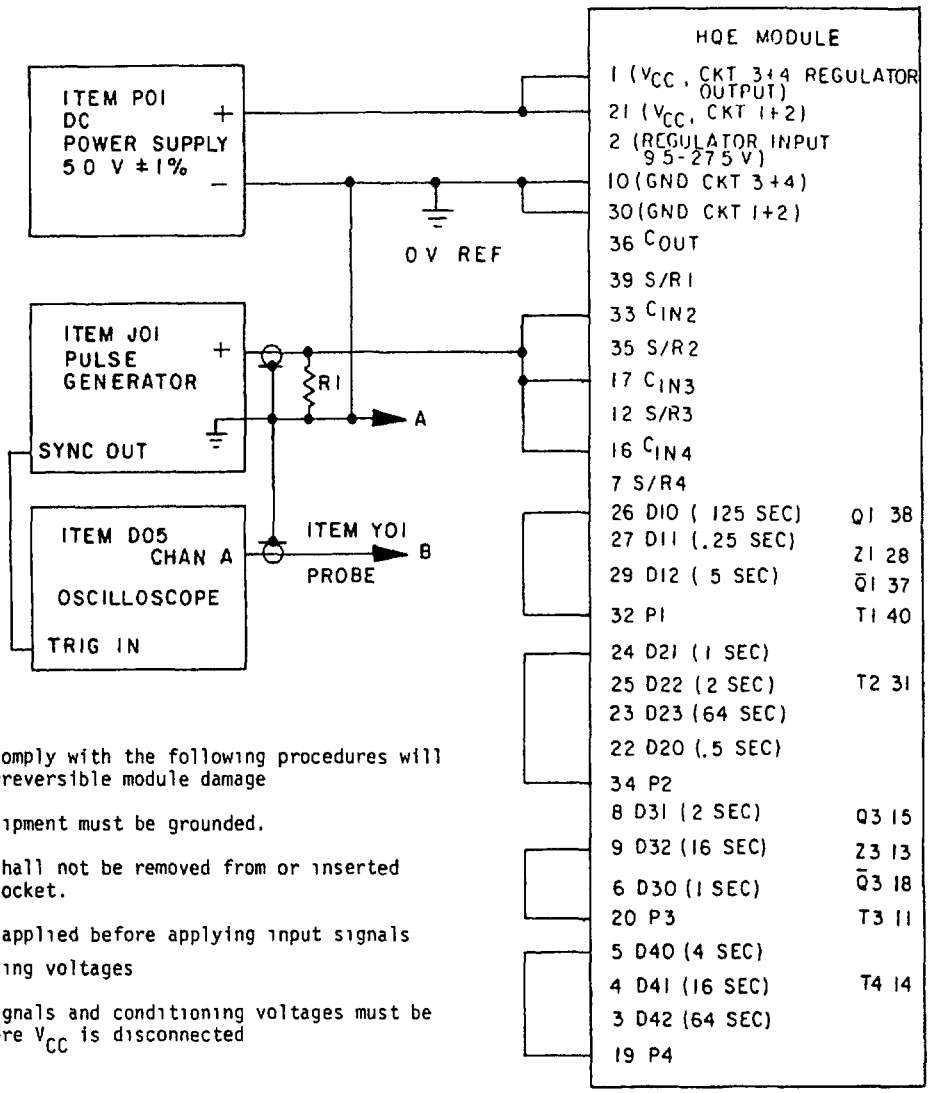
TABLE XXIII Timer 1-4 disable test

| Test condition | Parameter | Connect | | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| U1 | Timer 2 disable | 35 | 24 | 4.5 | 5.0 | V | 29, 30 |
| U2 | Timer 3 disable | 12 | 9 | ↓ | ↓ | ↓ | ↓ |
| U3 | Timer 4 disable | 7 | 5 | ↓ | ↓ | ↓ | ↓ |
| U4 | Timer 1 disable | 39 | 26 | 4.5 | 5.0 | V | 29, 30 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

$V_{CC}$ must be applied before applying input signals or conditioning voltages

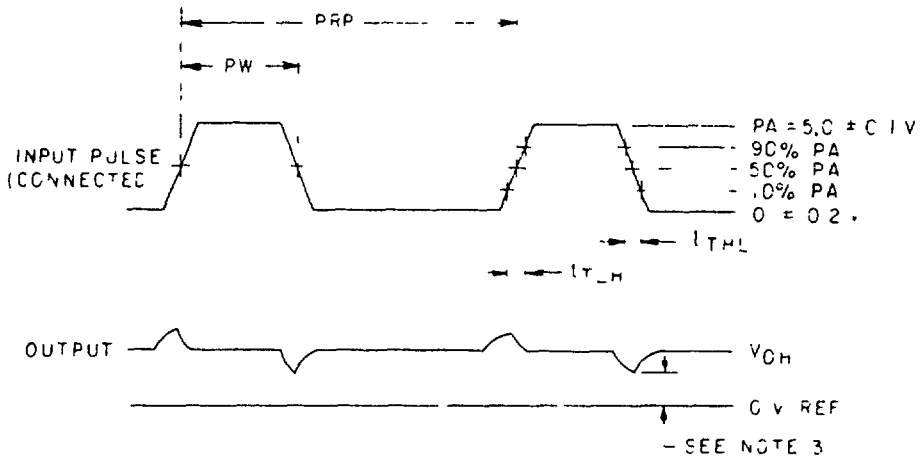
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTES
1 R1 = 51Ω ±5%, 1/4 W.
2 For test condition U1 - U3 measurement must be taken 22 µs after point A is connected, monitor steady state voltage for 33 µs.
3 For test condition U4 measurement must be taken 0.25 ms after point A is connected, monitor steady state voltage for 0 3 second

FIGURE 29 Timer 1-4 disable test circuit for table XXIII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1. Input pulse parameters
   PRP = 1.33 μs ± 5%
   PW = 0.66 μs ± 5%
   $t_{TLH}$, $t_{THL}$ = 15 ± 2 ns
2. All voltages are referenced to 0 V unless otherwise specified.
3. Make measurement from 0 V reference to the most negative portion of output waveform.

FIGURE 30. Timer 1-4 disable waveforms for table XXIII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXIV Timer 2-4 output transition time test

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| V1 | D20 timer transition time ($t_{THL}$) | 22 | 5 | 2 | 1,500 | ns | 31, 32 |
| V2 | D21 | 24 | 5 | | | | |
| V3 | D22 | 25 | 5 | | | | |
| V4 | D23 | 23 | 3 | | | | |
| V5 | D31 | 8 | 25 | | | | |
| V6 | D30 | 6 | 25 | | | | |
| V7 | D32 | 9 | 3 | | | | |
| V8 | D40 | 5 | 9 | | | | |
| V9 | D41 | 4 | 9 | | | | |
| V10 | D42 timer transition time ($t_{THL}$) | 3 | 23 | | | | |
| V11 | D20 timer transition time ($t_{TLH}$) | 22 | 5 | | | | |
| V12 | D21 | 24 | 5 | | | | |
| V13 | D22 | 25 | 5 | | | | |
| V14 | D23 | 23 | 3 | | | | |
| V15 | D31 | 8 | 25 | | | | |
| V16 | D30 | 6 | 25 | | | | |
| V17 | D32 timer transition time ($t_{TLH}$) | 9 | 3 | 2 | 1,500 | ns | 31, 32 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

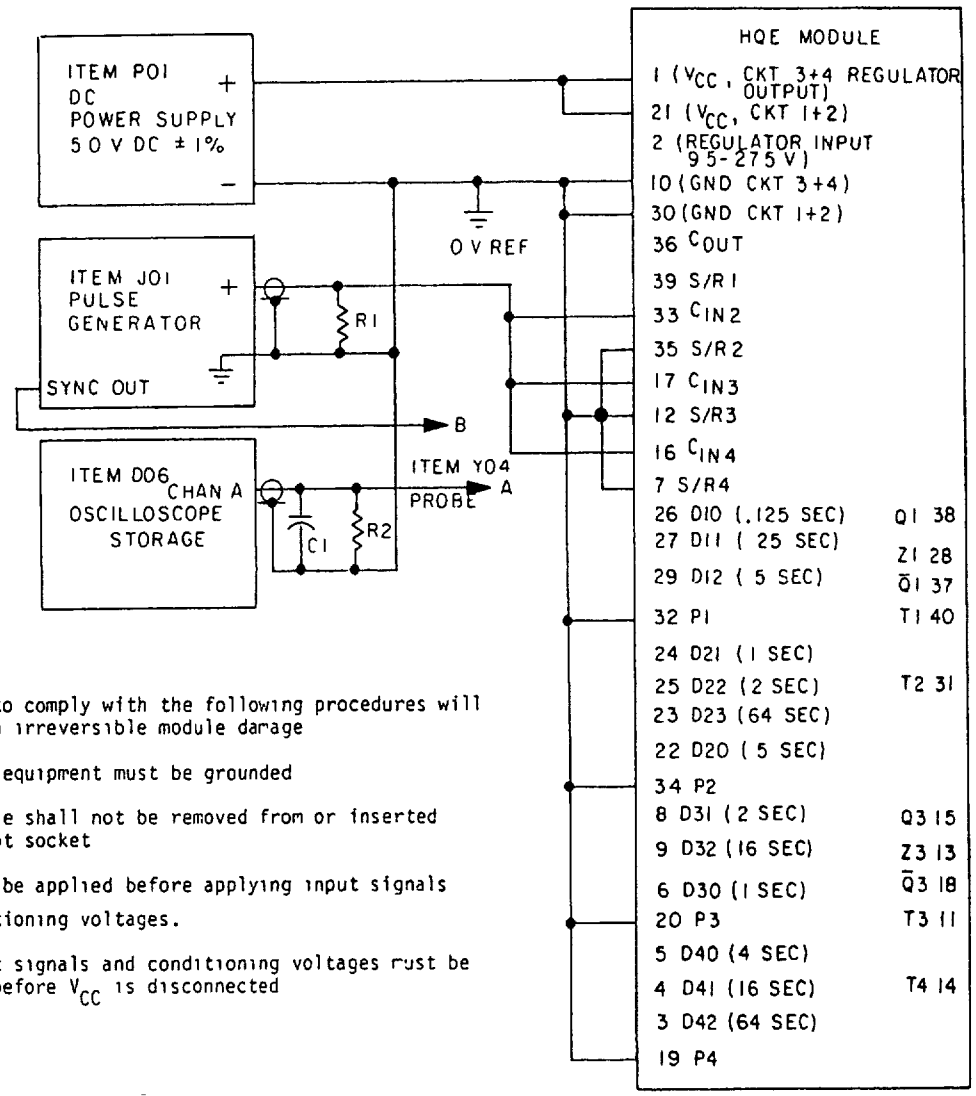
TABLE XXIV. Timer 2-4 output transition time test - Continued

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| V18 | D40 timer transition time ($t_{TLH}$) | 5 | 9 | 2 | 1,500 | ns | 31, 32 |
| V19 | D41 timer transition time ($t_{TLH}$) | 4 | 9 | 2 | 1,500 | ns | 31, 32 |
| V20 | D42 timer transition time ($t_{TLH}$) | 3 | 23 | 2 | 1,500 | ns | 31, 32 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION:

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages.

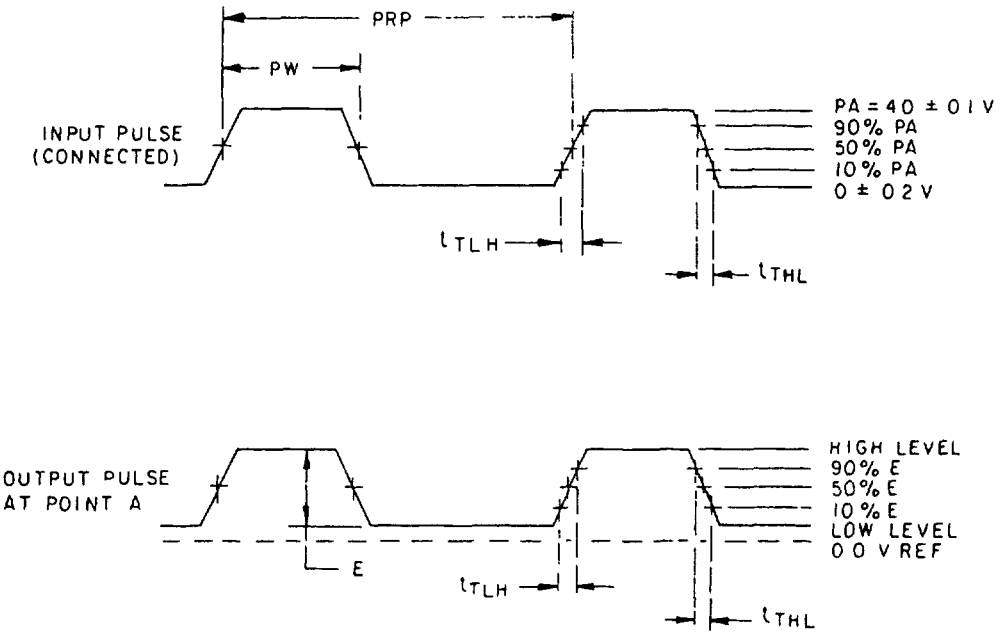
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTE R1 = 51Ω ±5%, 1/4 W
R2 = 200 kΩ ±1%, 1/8 W
C1 = 50 ±5 pF (including probe and fixture capacitance)

FIGURE 31 Timer 2-4 output transition test circuit for table XXIV



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1 Input pulse parameters
PRP = 1 333 µs ±5%
PW = 0 666 µs ±5%
$t_{TLH}$, $t_{THL}$ = 15 ±2 ns
2 Before making the measurement specified in table XXIV, verify that the output low level is ≤ 0 5 V and the high level is ≥ 4 2 V

FIGURE 32 Timer 2-4 output transition time waveforms for table XXIV



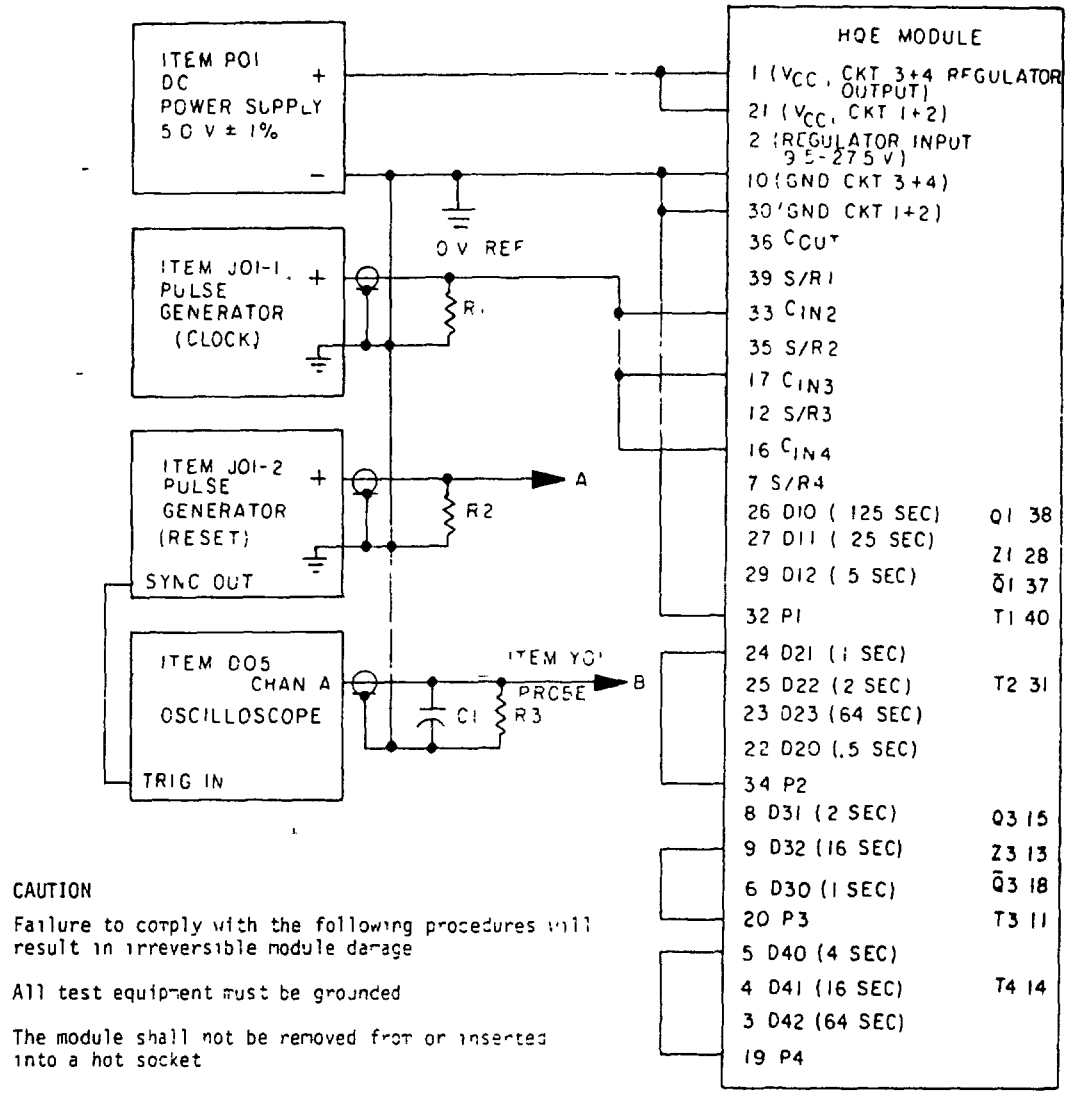
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXV. Timer 2-4 reset propagation delay time test

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| W1 | D20 reset propagation delay time ($t_{PHL}$) | 35 | 24 | 0.0 | 4.3 | µs | 33, 34 |
| W2 | D30 reset propagation delay time ($t_{PHL}$) | 12 | 9 | 0 0 | 4.3 | µs | 33, 34 |
| W3 | D40 reset propagation delay time ($t_{PHL}$) | 7 | 5 | 0.0 | 4.3 | µs | 33, 34 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

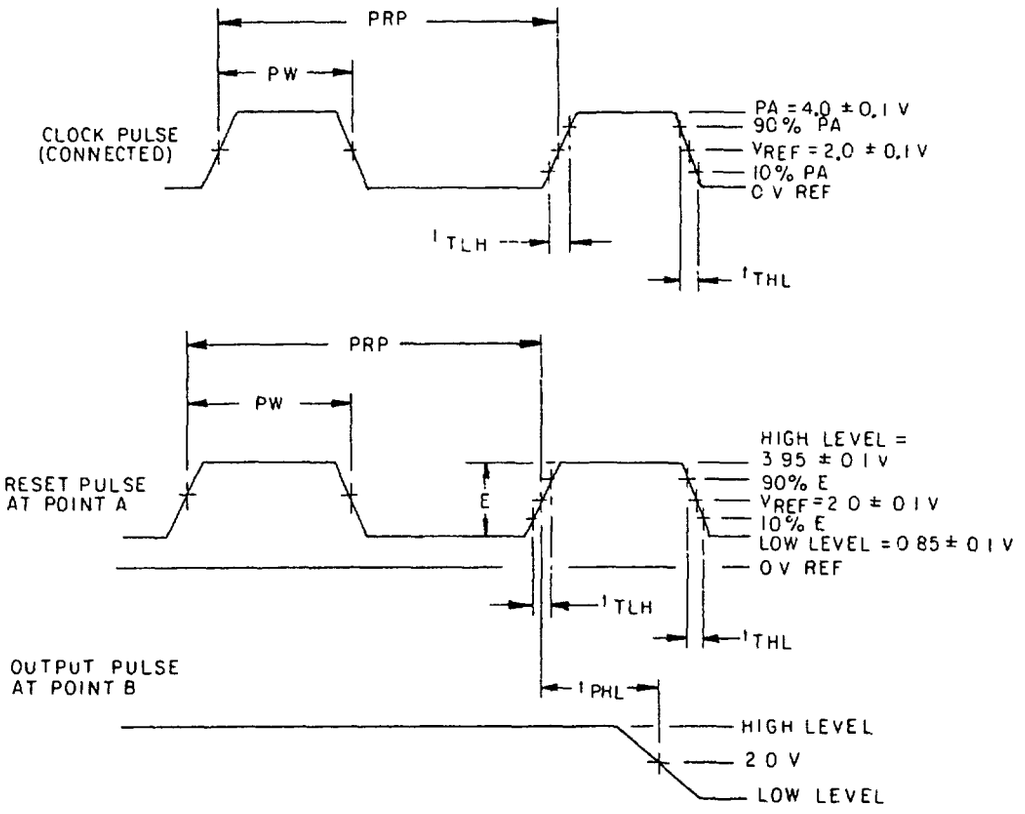
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTE R1 = R2 = 51Ω ±5%, 1/4 W.
R3 = 200 kΩ ±1%, 1/8 W
C1 = 50 ±5 pF (including probe and fixture capacitance)

FIGURE 33 Timer 2-4 reset propagation delay time ($t_{PLH}$) test circuit for table XXV



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1. Clock pulse parameters
   PRP = 1.33 ±0.1 µs
   PW = 0.670 ± .05 µs
   $t_{TLH}$, $t_{THL}$ = 15 ±2 ns
2. Reset pulse parameters
   PRP = 700 ±20 µs.
   PW = 2.8 +.2, -0 µs.
   $t_{TLH}$, $t_{THL}$ = 15 ±2 ns
3. Before making the measurements specified in table XXV verify that the output pulse high level is ≥ 4.2 V and low level is ≤ 0.4 V.
4. All voltages are referenced to 0 V unless otherwise specified

FIGURE 34 Timer 2-4 propagation delay time waveforms for table XXV



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

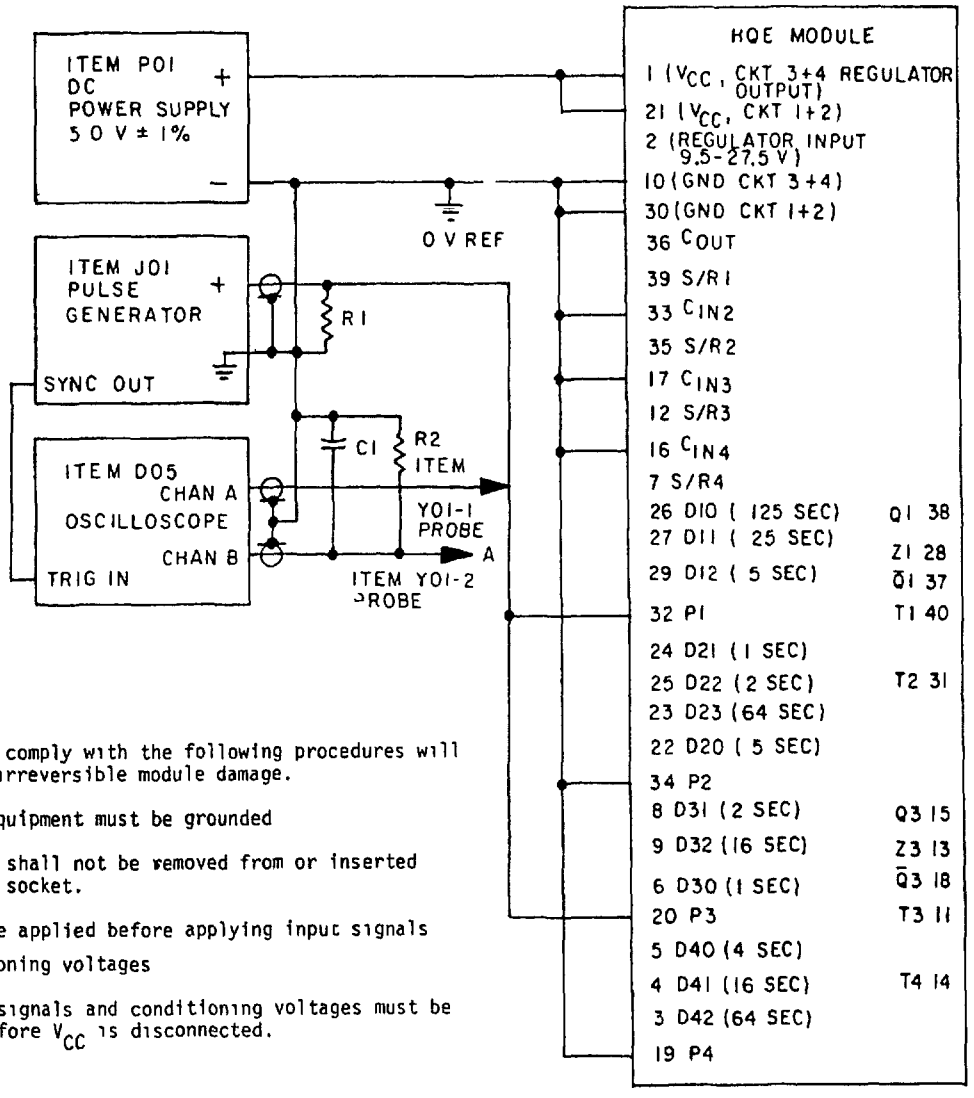
TABLE XXVI Q and $\bar{Q}$ output propagation delay time and transition time test.

| Test condition | Parameter | Connect | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| X1 | Q1 propagation delay time ($t_{PHL}$) | 38 | 2 | 580 | ns | 35, 36 |
| X2 | Q3 propagation delay time ($t_{PHL}$) | 15 | | 580 | | |
| X3 | $\bar{Q}$1 propagation delay time ($t_{PLH}$) | 37 | | 435 | | |
| X4 | $\bar{Q}$3 propagation delay time ($t_{PLH}$) | 18 | | 435 | | |
| X5 | Q1 propagation delay time ($t_{PLH}$) | 38 | | 430 | | |
| X6 | Q3 propagation delay time ($t_{PLH}$) | 15 | | 430 | | |
| X7 | $\bar{Q}$1 propagation delay time ($t_{PHL}$) | 37 | | 370 | | |
| X8 | $\bar{Q}$3 propagation delay time ($t_{PHL}$) | 18 | | 370 | | |
| X9 | Q1 transition time ($t_{THL}$) | 38 | | 85 | | |
| X10 | $\bar{Q}$1 transition time ($t_{THL}$) | 37 | | | | |
| X11 | Q3 transition time ($t_{THL}$) | 15 | | | | |
| X12 | $\bar{Q}$3 transition time ($t_{THL}$) | 18 | | 85 | | |
| X13 | Q1 transition time ($t_{TLH}$) | 38 | | 390 | | |
| X14 | $\bar{Q}$1 transition time ($t_{TLH}$) | 37 | | | | |
| X15 | Q3 transition time ($t_{TLH}$) | 15 | | | | |
| X16 | $\bar{Q}$3 transition time ($t_{TLH}$) | 18 | 2 | 390 | ns | 35, 36 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION:

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket.

$V_{CC}$ must be applied before applying input signals or conditioning voltages

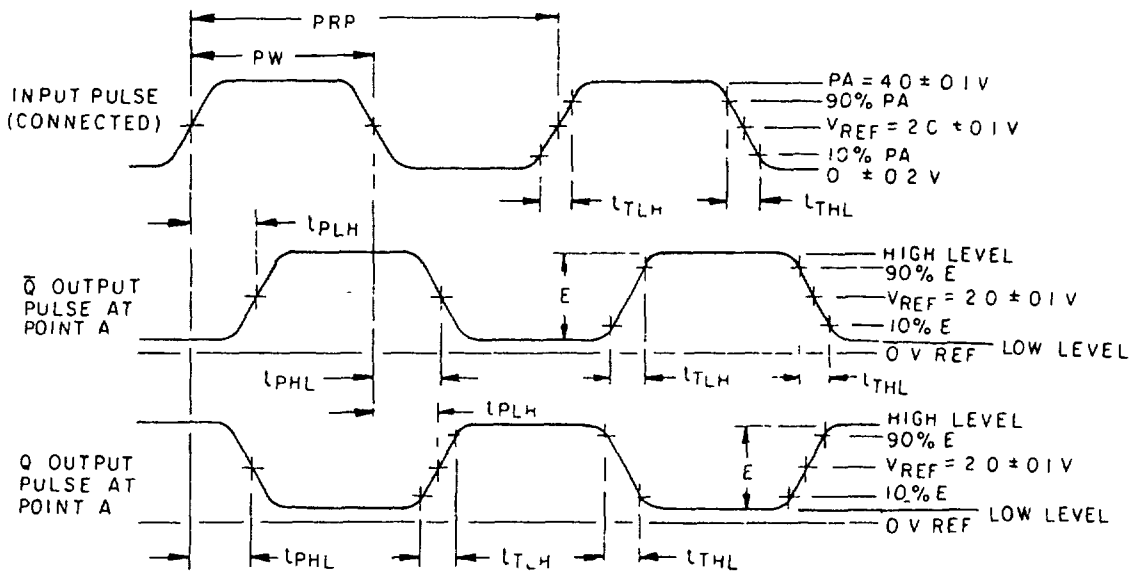
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected.

NOTE. R1 = 51Ω ±5%, 1/4 W.
R2 = 200 kΩ ±1%, 1/8 W.
C1 = 50 ±5 pF (including probe and fixture capacitance).

FIGURE 35 Q and $\overline{Q}$ output propagation delay and transition time test circuit for table XXVI



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1 Input pulse parameters
PRP = 100 ±10 µs
PW = 50 ±5 µs.
$t_{TLH}$, $t_{THL}$ = 10 ±2 ns
2 Before making the measurements specified in table XXV, verify that the output high level pulse level is ≥ 4 2 V and the low level is ≤ 0 5 V
3 All voltages are referenced to 0 V unless otherwise specified

FIGURE 36 Q and $\overline{Q}$ output propagation delay and transition time waveforms for table XXVI



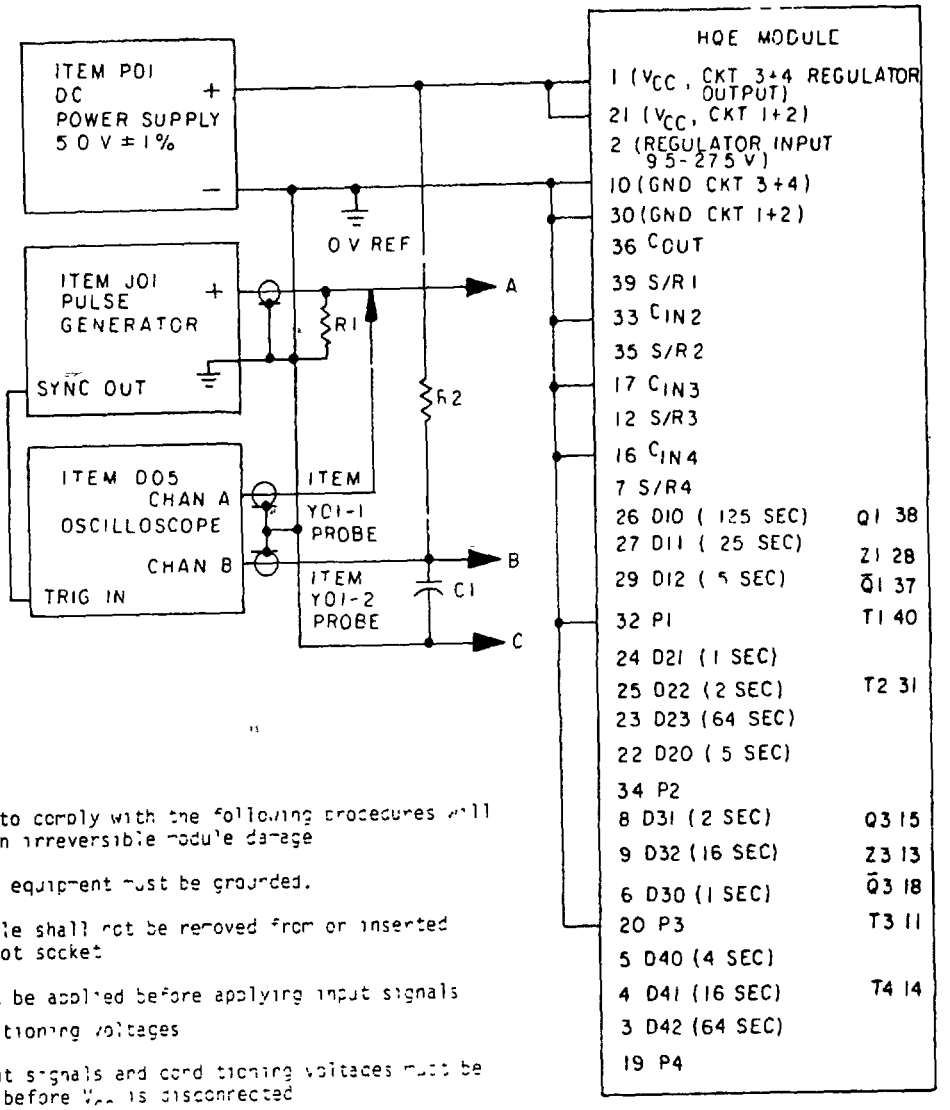
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXVII T1-T4 output propagation delay time and transition time test.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| Y1 | T1 propagation delay time ($t_{PHL}$) | 28 | 40 | 34, 19 | 2 | 200 | ns | 37, 38 |
| Y2 | T3 propagation delay time ($t_{PHL}$) | 13 | 11 | 34, 19 | | 200 | | |
| Y3 | T1 propagation delay time ($t_{PLH}$) | 28 | 40 | 34, 19 | | 1,500 | | |
| Y4 | T3 propagation delay time ($t_{PLH}$) | 13 | 11 | 34, 19 | | 1,500 | | |
| Y5 | T2 propagation delay time ($t_{PLH}$) | 34 | 31 | 19 | | 2,100 | | |
| Y6 | T4 propagation delay time ($t_{PLH}$) | 19 | 14 | 34 | | 1,900 | | |
| Y7 | T2 propagation delay time ($t_{PHL}$) | 34 | 31 | 19 | | 720 | | |
| Y8 | T4 propagation delay time ($t_{PHL}$) | 19 | 14 | 34 | | 610 | | |
| Y9 | T1 transition time ($t_{THL}$) | 28 | 40 | 34, 19 | | 200 | | |
| Y10 | T3 transition time ($t_{THL}$) | 13 | 11 | 34, 19 | | 200 | | |
| Y11 | T1 transition time ($t_{TLH}$) | 28 | 40 | 34, 19 | | 400 | | |
| Y12 | T3 transition time ($t_{TLH}$) | 13 | 11 | 34, 19 | | 400 | | |
| Y13 | T2 transition time ($t_{THL}$) | 34 | 31 | 19 | | 200 | | |
| Y14 | T4 transition time ($t_{THL}$) | 19 | 14 | 34 | | 200 | | |
| Y15 | T2 transition time ($t_{TLH}$) | 34 | 31 | 19 | | 400 | | |
| Y16 | T4 transition time ($t_{TLH}$) | 19 | 14 | 34 | 2 | 400 | ns | 37, 38 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

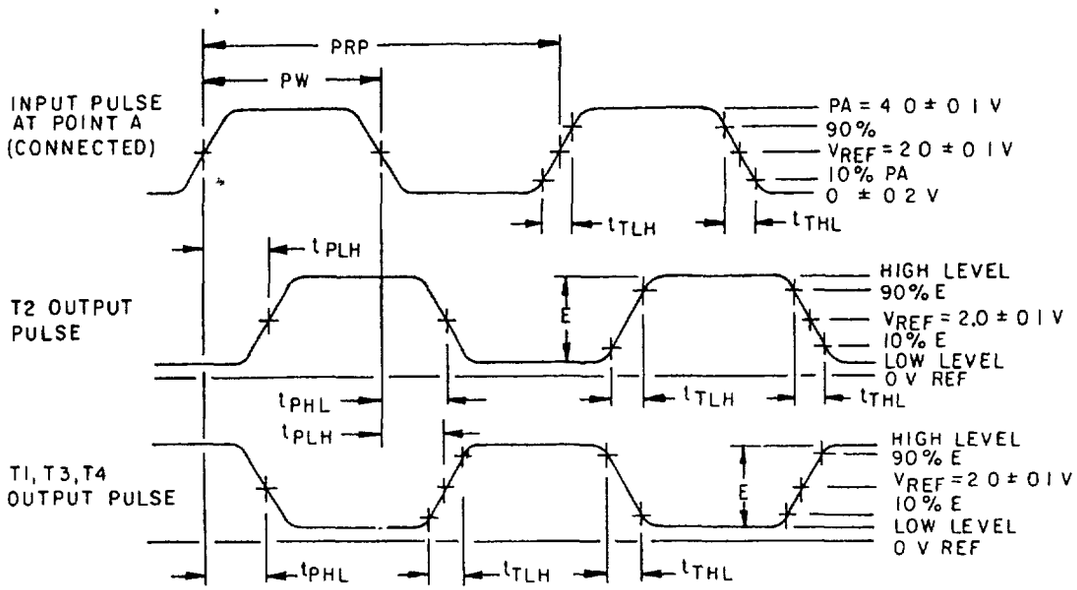
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTE R1 = 51Ω ±5%, 1/4 W
R2 = 187Ω ±1%, 1/4 W
C1 = 50 ±5 pF (including probe and fixture capacitance)

FIGURE 37 T1 - T4 output propagation delay and transition time test for table XXVII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1 Input pulse parameters
PRP = 100 ±10 µs
PW = 50 ±5 µs
$t_{TLH}$, $t_{THL}$ = 10 ±2 ns
2 Before making the measurement specified in table XXVII, verify that the output pulse high level is $\geq$ 2 4 V and the low level is $\leq$ 0 5 V
3 All voltages are referenced to 0 V unless otherwise specified

FIGURE 38 T1 - T4 output propagation delay and transition time waveforms for table XXVII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXVIII Output crosstalk, low-level and high-level test.

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| Z1 | D10 output crosstalk, low level | 38 to 28<br>15 to 13 | 34,20, 19,32 | 26 | 35,12, 7 | Open | 0 | 400 | mV | 39, 40 |
| Z2 | D20 | 38 to 28<br>15 to 13 | 34,20, 19,32 | 22 | 39,12, 7 | | | | | |
| Z3 | D30 | 38 to 28<br>15 to 13 | 34,20, 19,32 | 9 | 39,35, 7 | | | | | |
| Z4 | D40 | 38 to 28<br>15 to 13 | 34,20, 19,32 | 5 | 39,35, 12 | | | | | |
| Z5 | Q1 | 15 to 13<br>28 to 38 | 34,20, 19 | 37 | 39,35, 7,12, 32 | Open | | | | |
| Z6 | $\bar{Q}1$ output crosstalk, low level | 15 to 13<br>37 to 28 | 34,20, 19 | 38 | 39,35, 7,12 | 32 | 0 | 400 | mV | 39, 40 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXVIII Output crosstalk, low-level and high-level test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| Z7 | Q3 output crosstalk, low level | 38 to 28 15 to 13 | 32,34, 19 | 18 | 39,35, 20,7, 12 | Open | 0 | 400 | mV | 39, 40 |
| Z8 | $\bar{Q}3$ | 38 to 28 18 to 13 | 32,34, 19 | 15 | 39,35, 7,12 | 20 | | | | |
| Z9 | T1 | 15 to 13 28 to 38 | 34,20, 19 | 40 | 35,39, 12,7, 32 | Open | | | | |
| Z10 | T2 | | 32,19, 20 | 31 | 35,39, 12,7, 34 | Open | | | | |
| Z11 | T3 | | 32,34, 19 | 11 | 35,39, 12,7, 20 | Open | | | | |
| Z12 | T4 output crosstalk, low level | 15 to 13 28 to 38 | 32,20, 34 | 14 | 35,39, 7,12 | 19 | 0 | 400 | mV | 39, 40 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXVIII. Output crosstalk, low-level and high-level test Continued

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test Figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| Z13 | D10 output crosstalk, high level | 38 to 28<br>15 to 13<br>26 to 32 | 34,20, 19 | 26 | 35,39, 7,12 | Open | 0 | 400 | mV | 39, 40 |
| Z14 | D20 | 38 to 28<br>15 to 13<br>22 to 34 | 32,20, 19 | 22 | 35,39, 7,12 | | | | | |
| Z15 | D30 | 38 to 28<br>15 to 13<br>20 to 9 | 32,34, 19 | 9 | 35,39, 7,12 | | | | | |
| Z16 | D40 | 38 to 28<br>15 to 13<br>5 to 19 | 32,34, 20 | 5 | 35,39, 7,12 | | | | | |
| Z17 | $\bar{Q}1$ | 15 to 13<br>28 to 37 | 34,20, 19 | 38 | 39,35, 12,7, 32 | Open | | | | |
| Z18 | Q1 output crosstalk, high level | 15 to 13<br>38 to 28 | 34,20, 19 | 37 | 39,35, 12,7 | 32 | 0 | 400 | mV | 39, 40 |





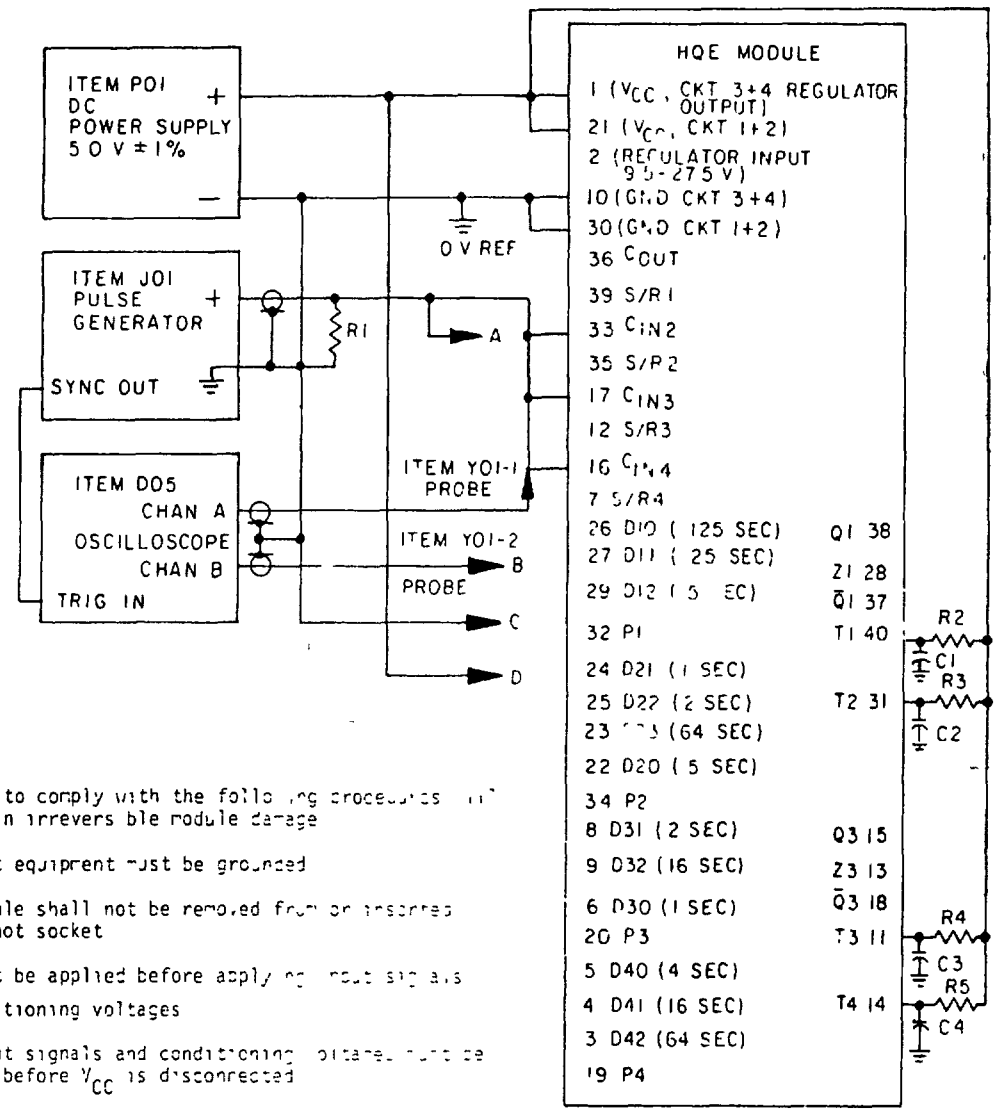
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXVIII Output crosstalk, low-level and high-level test - Continued

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin to pin | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| Z19 | $\overline{Q}$3 output crosstalk, high level | 38 to 28<br>15 to 13 | 19,32, 34 | 15 | 39,35, 12,7, 20 | Open | 0 | 400 | mV | 39, 40 |
| Z20 | Q3 | 37 to 28<br>13 to 15 | 19,32, 34 | 18 | 39,35, 12,7 | 20 | | | | |
| Z21 | T1 | 38 to 28<br>15 to 13 | 34,20, 19 | 40 | 39,35, 12,7 | 32 | | | | |
| Z22 | T2 | 38 to 28<br>15 to 13 | 32,20 19 | 31 | 39,35, 12,7 | 34 | | | | |
| Z23 | T3 | 38 to 28<br>18 to 13 | 34,32, 19 | 11 | 39,35, 12,7, 20 | Open | | | | |
| Z24 | T4 output crosstalk, high level | 38 to 28<br>15 to 13 | 34,20, 32 | 14 | 39,35, 12,7, 19 | Open | 0 | 400 | mV | 39, 40 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages

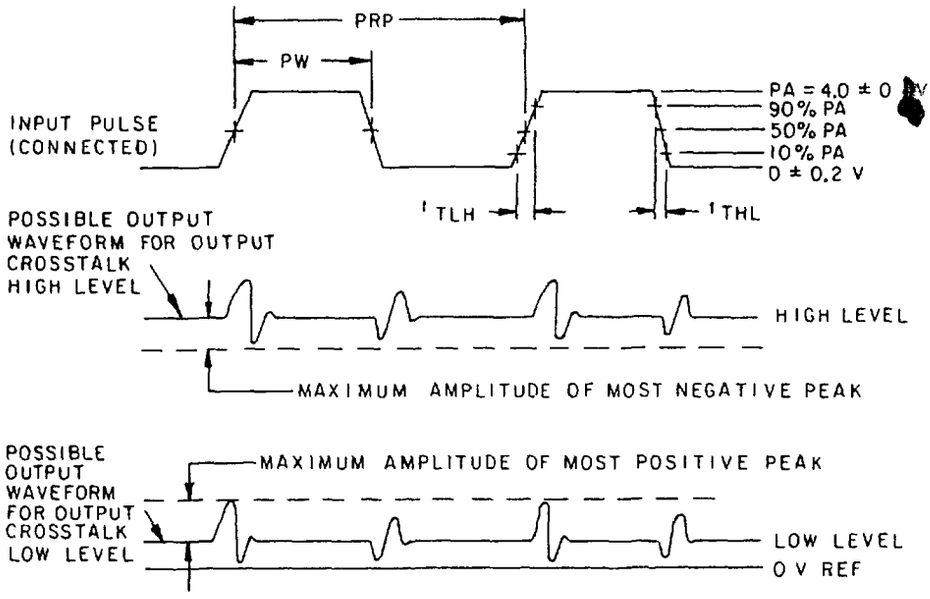
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTE R1 = 51Ω ±5%, 1/4 W
R2 - R5 = 187Ω ±1%, 1/4 W
C1 - C4 = 50 ±5 pF (including probe and fixture capacitance).

FIGURE 39 Output crosstalk test circuit for table XXVIII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1. Input pulse parameters
   PRP = 50 µs ±5%
   PW = 25 µs ±5%
   $t_{TLH}$, $t_{THL}$ = 15 ±2 ns
2 All voltages are referenced to 0 V, unless otherwise specified

FIGURE 40 Output crosstalk low level and high level waveforms for table XXVIII



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

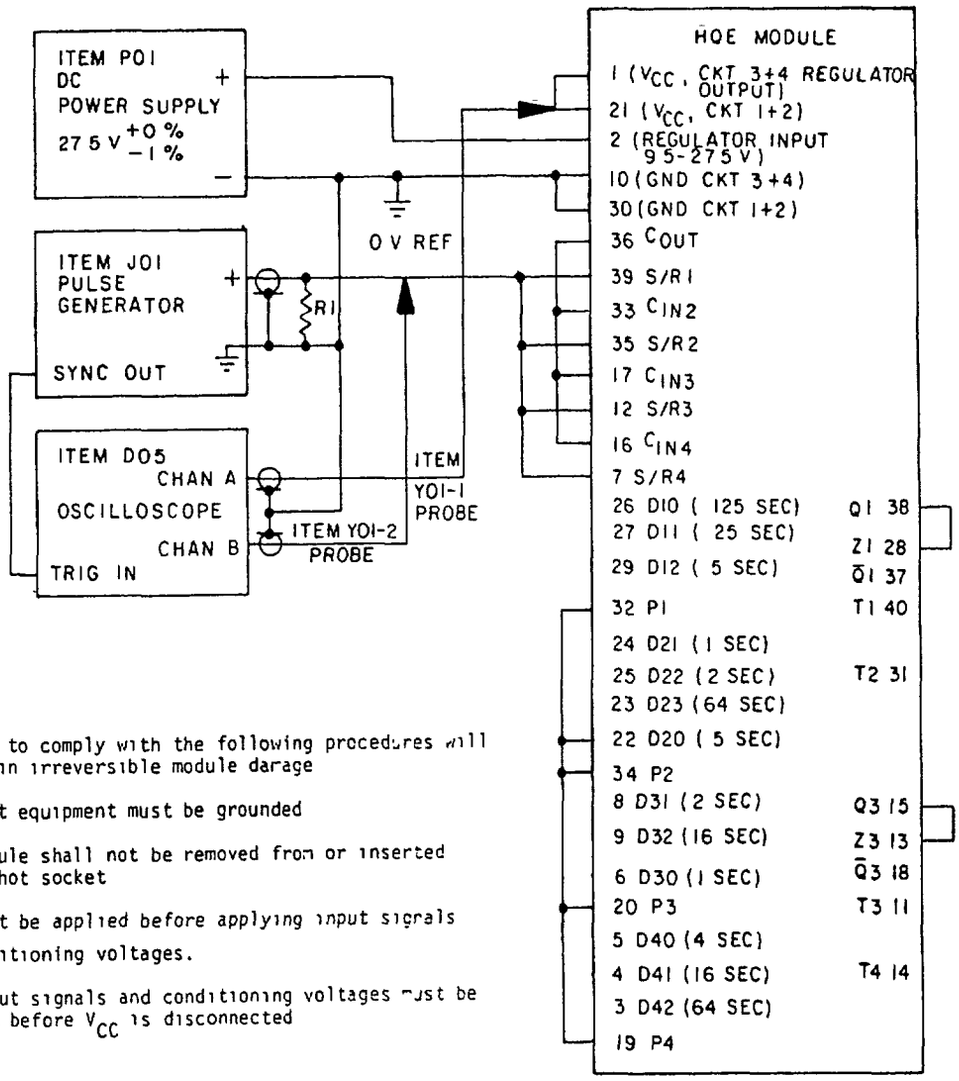
TABLE XXIX Power supply transient amplitude test

| Test condition | Parameter | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| AA | Power supply transient amplitude test | 0 | 500 | mV | 41, 42 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltages.

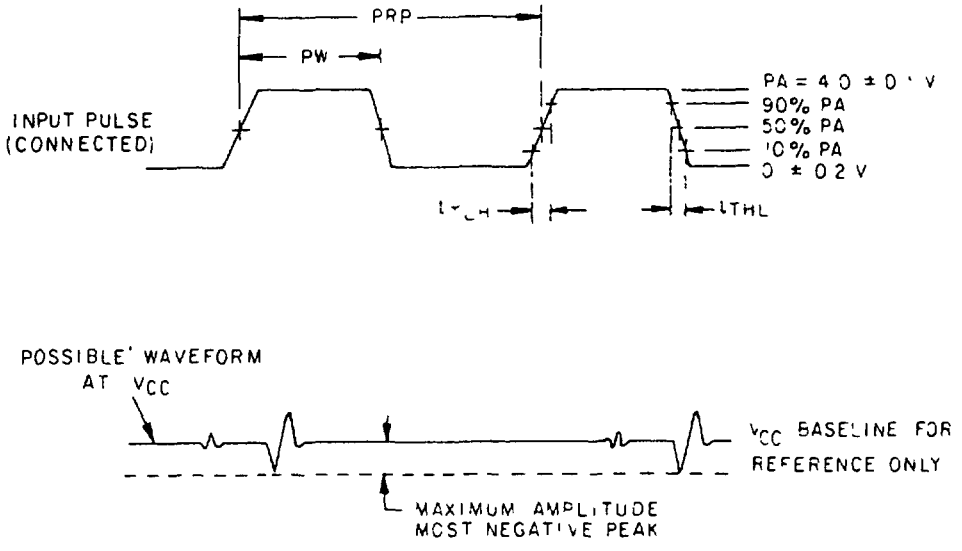
All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTE R1 = 51Ω ±5%, 1/4 W

FIGURE 41 Power supply transient amplitude test circuit for table XXIX



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




NOTES
1 Input pulse parameters
PRP = 1.3 µs ±5%
PW = 667 ns ±5%
$t_{TLH}$, $t_{THL}$ = 15 ±2 ns
2 All voltages are referenced to 0 V, unless otherwise specified

FIGURE 42 Power supply transient amplitude waveforms for table XXIX



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXX Power supply filter capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| BB1 | Power supply filter capacitance | Open | 1 | 0 01 | 0 04 | µF | 43 |
| BB2 | Power supply filter capacitance | Open | 21 | 0 01 | 0.04 | µF | 43 |





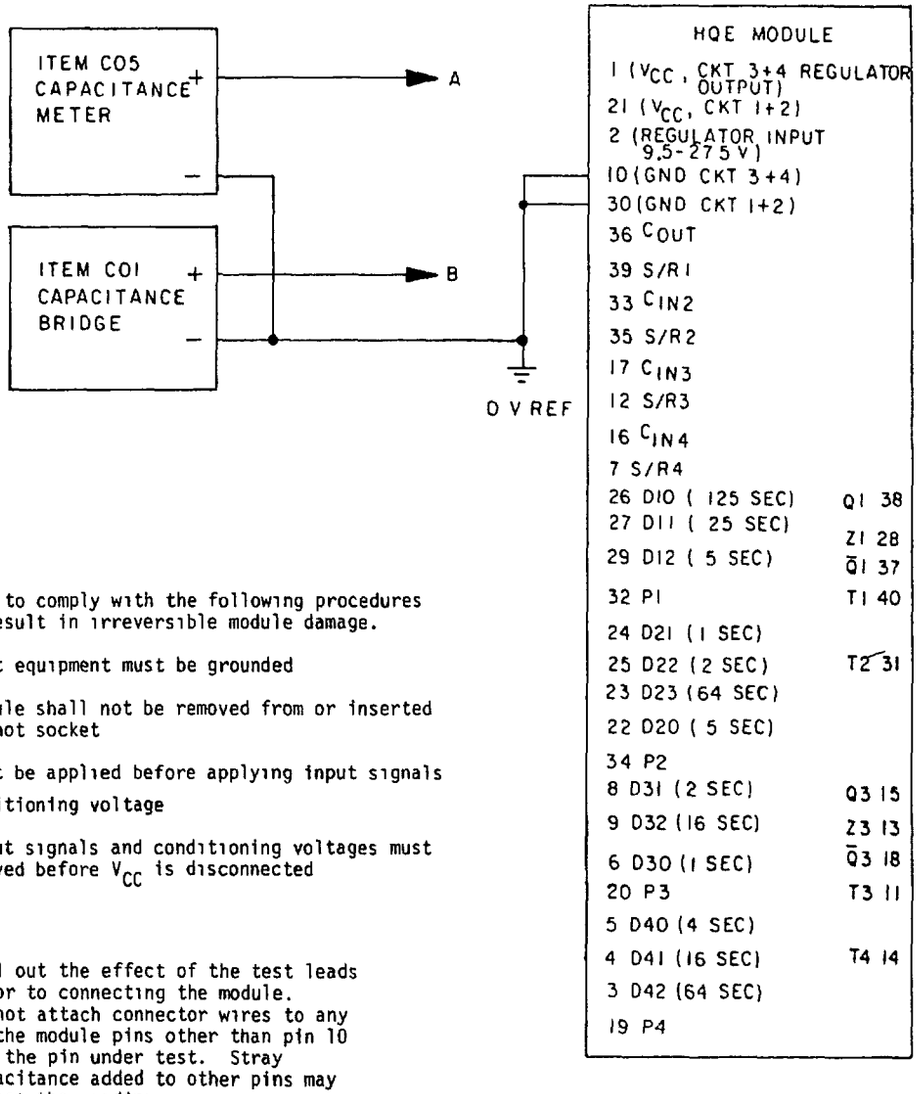
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

TABLE XXXI Input capacitance test.

| Test condition | Parameter | Cond 1 A to pin | Cond 1 B to pin | Initial limits +25°C Ambient Min | Initial limits +25°C Ambient Max | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| CC1 | Input capacitance | 33 | Open | 2 | 12 | pF | 43 |
| CC2 | | 17 | | | 12 | | |
| CC3 | | 16 | | | 12 | | |
| CC4 | | 32 | | | 20 | | |
| CC5 | | 34 | | | | | |
| CC6 | | 20 | | | | | |
| CC7 | | 19 | | | | | |
| CC8 | | 39 | | | | | |
| CC9 | | 35 | | | | | |
| CC10 | | 12 | | | | | |
| CC11 | | 7 | | | 20 | | |
| CC12 | | 28 | | | 40 | | |
| CC13 | Input capacitance | 13 | Open | 2 | 40 | pf | 43 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.




CAUTION

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded

The module shall not be removed from or inserted into a hot socket

$V_{CC}$ must be applied before applying input signals or conditioning voltage

All input signals and conditioning voltages must be removed before $V_{CC}$ is disconnected

NOTES
1 Null out the effect of the test leads prior to connecting the module.
2 Do not attach connector wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
3 Capacitance bridge (COI) set up
Frequency = 1 kHz
Vpk = 0.3 V maximum

FIGURE 43 Input and filter capacitance test circuit for tables XXX and XXXI



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

# IDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

RUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/208

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID RESTRICTIVE LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:40Z
Check the source to verify that this is the current version before use.